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TRAFFIC SIGNAL NOTES

1. INSTALL SIGNALS HORIZONTALLY ON MAST ARM, 17 FT.-6 IN. ABOVE THE ROADWAY.
2. FURNISH BLACK HOUSING FOR VEHICLE AND PEDESTRIAN SIGNALS. FURNISH BLACK VEHICLE SIGNAL HEAD BACK PLATES WITH 2 IN. RETROFLECTIVE YELLOW BORDER.
3. FURNISH VEHICLE AND PEDESTRIAN SIGNALS WITH LIGHT EMITTING DIODE (LED) SIGNAL LAMP UNITS.
4. USE TYPE B (HIGH INTENSITY PRISMATIC) OR TYPE D (DIAMOND GRADE) RETROREFLECTIVE SHEETING FOR SIGNS MOUNTED UNDER OR ADJACENT TO THE SIGNAL HEADS.
5. FURNISH SYMBOL TYPE PEDESTRIAN COUNTDOWN SIGNALS. INSTALL USING MOUNTING HEIGHT IN ACCORDANCE WITH THE LATEST TEXAS MANUAL ON UNIFORM TRAFFIC CONTROL DEVICES.
6. FURNISH MATERIALS NECESSARY TO INSTALL ACCESSIBLE PEDESTRIAN SIGNAL UNITS AND SIGNS AS SHOWN IN THE PLANS. INSTALL AT 3 FT.-6 IN. TO 4 FT.-0 IN. ABOVE THE SIDEWALK OR CONCRETE WALKWAY.
7. ROUTE CABLE FOR LUMINAIRES (#12/4C-TRAY CABLE) TO THE SERVICE ENCLOSURE. SEE ELECTRICAL DETAIL SHEETS. DO NOT PASS LUMINAIRE CONDUCTORS THROUGH THE SIGNAL CONTROLLER CABINET.
8. FURNISH AND INSTALL FULL-ACTUATED CONTROLLER WITH INTERNAL TIME BASE COORDINATION UNIT IN A CABINET, MOUNTED ON AN 18-INCH BASE EXTENSION.
9. FURNISH ALL MATERIALS. SUPPLY THE CONTROLLER WITH DETECTION PHASE SEQUENCE, DETECTOR UNITS, DETECTOR CARDS, DETECTOR CARD RACK, AND POWER SUPPLY, TO THE DEPARTMENT'S SIGNAL SHOP, 6810 KATY ROAD, HOUSTON, TEXAS FORTY-FIVE (45) DAYS IN ADVANCE FOR INSPECTION, SET UP, AND TESTING. CONTACT MR. EUGENE AMPOMAH, P.E., IN WRITING, AT LEAST FIFTEEN (15) WORKING DAYS PRIOR TO PICKING UP THE MATERIALS.

   ADDRESS: TEXAS DEPARTMENT OF TRANSPORTATION
   P.O. BOX 1386
   HOUSTON, TEXAS 77251-1386
   TEL.NO. (713) 802-5378

10. THE DEPARTMENT'S TRAFFIC SIGNAL MAINTENANCE OFFICE WILL PROVIDE PHASING AND TIMING FOR PERMANENT TRAFFIC SIGNALS.
11. LOCATE CABINET(S), STEEL SIGNAL POLES, SIGNAL DETECTORS, ETC., AS APPROVED.
12. REPAIR OR REPLACE PAVEMENT AND SIDEWALKS DAMAGED BY THE CONTRACTOR'S FORCES DURING CONSTRUCTION AT NO COST TO THE DEPARTMENT.
13. ALL TRAFFIC SIGNAL DETECTION DEVICES AND RELATED COMPONENTS SHALL BE SALVAGED AND RETURNED TO THE DEPARTMENT'S SIGNAL SHOP AT 6810 OLD KATY ROAD, HOUSTON, TEXAS, BETWEEN 9:00 AM AND 3:00 PM, MONDAY THROUGH FRIDAY. CAREFULLY REMOVE THE MATERIALS SO THAT THEY WILL NOT BE MARRED OR DAMAGED. REPLACE MATERIALS THAT ARE SCARRED, BATTERED OR BROKEN BY THE CONTRACTOR AT NO EXPENSE TO THE DEPARTMENT.
14. FOR ALL OTHER TRAFFIC SIGNAL-RELATED COMPONENTS, CONTACT MR. EUGENE AMPOMAH, P.E., AT TEXAS DEPARTMENT OF TRANSPORTATION, P.O.BOX 1386, HOUSTON, TEXAS 77251-1386, TEL. NO. (713) 802-5378; HIS EMPLOYEES WILL DETERMINE WHICH ITEMS WILL BE SALVAGED. ITEMS DEEMED SALVAGEABLE WILL BE DELIVERED TO THE DEPARTMENT'S SIGNAL SHOP AT 6810 OLD KATY ROAD, HOUSTON, TEXAS, BETWEEN 9:00 AM AND 3:00 PM, MONDAY THROUGH FRIDAY. CAREFULLY REMOVE THE MATERIALS SO THAT THEY WILL NOT BE MARRED OR DAMAGED. REPLACE MATERIALS THAT ARE SCARRED, BATTERED OR BROKEN BY THE CONTRACTOR AT NO EXPENSE TO THE DEPARTMENT. DISPOSE OF OTHER ITEMS REMOVED BY THE CONTRACTOR AT NO EXPENSE TO THE DEPARTMENT.
15. ASSUME OWNERSHIP OF THE REMOVED EXISTING SIGNS.
16. SEAL ENDS OF ALL CONDUITS WITH DUCT SEAL, EXPANDABLE FOAM, OR BY OTHER METHODS APPROVED BY THE ENGINEER. SEAL CONDUIT IMMEDIATELY AFTER COMPLETION OF CONDUCTOR INSTALLATION AND PULL TESTS. DO NOT USE DUCT TAPE AS PERMANENT CONDUIT SEALANT. DO NOT USE SILICON CAULK AS A CONDUIT SEALANT.
17. CAP SPARE CONDUITS INSTALLED IN POLE FOUNDATIONS AND GROUND BOXES USING APPROVED CAPPING DEVICES.
18. DO NOT PLACE SIGNAL HEADS OVER THE ROADWAY UNTIL ALL NECESSARY MATERIALS ARE ON HAND AS APPROVED.
19. INSTALL TWO SET SCREWS ON ALL VEHICLE SIGNAL HEAD MOUNTING HARDWARE FITTINGS.
20. PROVIDE CONTINUED OPERATION OF THE EXISTING SIGNAL(S) DURING CONSTRUCTION AND UNTIL THE PROPOSED OPERATION IS COMPLETED.
21. ONCE THE INTEGRITY AND/OR FUNCTION OF THE EXISTING TRAFFIC SIGNAL(S) IS ALTERED BY THE CONTRACTOR, MAINTAIN AND OPERATE THE EXISTING TRAFFIC SIGNAL(S) UNTIL THE TRAFFIC SIGNAL WORK IS ACCEPTED BY THE DEPARTMENT. DURING THE CONSTRUCTION OF THE PROPOSED TRAFFIC SIGNAL WORK, MAINTAIN THE EXISTING TRAFFIC SIGNAL(S) AND/OR TEMPORARY CONSTRUCTION TRAFFIC SIGNAL(S) IN CONFORMANCE WITH THE LATEST TEXAS MANUAL ON UNIFORM TRAFFIC CONTROL DEVICES.
22. DURING CONSTRUCTION OF THE PROPOSED SIGNAL WORK, IF THE EXISTING TRAFFIC SIGNAL EQUIPMENT REQUIRES REPLACEMENT DUE TO WEAR, DETERIORATION, OR ANY CIRCUMSTANCE OVER WHICH THE CONTRACTOR HAS NO CONTROL, THE EQUIPMENT WILL BE FURNISHED BY THE DEPARTMENT AT NO COST TO THE CONTRACTOR. INSTALL THIS EQUIPMENT AT NO COST TO THE DEPARTMENT. SUCH MATERIALS WILL BE PROVIDED AT THE DEPARTMENT'S SIGNAL SHOP LOCATED AT 6810 KATY ROAD, HOUSTON, TEXAS. CONTACT MR. EUGENE AMPOMAH, P.E., AT TELEPHONE NUMBER (713) 802-5378.
23. MAINTAIN THE INTEGRITY AND FUNCTION OF EACH EXISTING SIGNALIZED INTERSECTION. ONCE THE INTEGRITY OR FUNCTION OF THE SIGNAL HAS BEEN ALTERED, PURSUE THE WORK AT THAT LOCATION WITHOUT DELAY OR INTERRUPTION TO RESTORE OPERATION TO ITS ORIGINAL OR FINAL OPERATIONAL DESIGN.
24. INSTALL A 5/8-IN. (MINIMUM) EYE BOLT FOR THE POINT OF ATTACHMENT BELOW THE SERVICE ENTRANCE WEATHERHEAD FOR THE SERVICE DROP TO STEEL OR WOOD POLE.
25. AIM LUMINAIRE ARMS MOUNTED ON TRAFFIC SIGNAL POLES PERPENDICULAR TO THE CENTERLINE OF THE ROADWAY IT IS INTENDED TO COVER, TO DEVELOP THE PROPER ILLUMINATION PATTERN FOR THE INTERSECTION.
26. PROVIDE 250-WATT-HPS (HIGH PRESSURE SODIUM) EQUIVALENT LIGHT EMITTING DIODE (LED) LUMINAIRES OPERATING AT 240 VOLTS.
27. WRAP SIGNAL HEADS WITH DARK PLASTIC OR SUITABLE MATERIAL TO CONCEAL THE SIGNAL FACES FROM THE TIME OF INSTALLATION UNTIL PLACING INTO OPERATION.
28. GROUND STEEL MAST ARM POLE ASSEMBLIES IN ACCORDANCE WITH REQUIREMENTS SHOWN ON THE LATEST TRAFFIC SIGNAL POLE FOUNDATION STANDARD. USE THE GROUNDING LUG ON THE POLE TO GROUND THE POLE TO THE GROUND CONDUCTORS FROM THE CONDUITS.
29. THE CONTRACTOR SHALL VERIFY PROPOSED SIGNAL POLE LOCATIONS AGAINST EXISTING UTILITIES PRIOR TO ORDERING THE POLES.
30. INSTALL A CLOSE NIPPLE WITH LOCK NUT AND BUSHING (SIZE AS REQUIRED) WHERE THE CABLE ENTERS THE UPPER PORTION OF THE SIGNAL POLE.
31. REFER TO TXDOT'S WEBSITE FOR PREQUALIFIED PRODUCTS LIST REGARDING RADAR DETECTORS, VIVDS CAMERAS, WIRELESS MAGNETOMETERS, VEHICLE LED TRAFFIC SIGNAL LAMP UNIT, SYMBOLIC PEDESTRIAN SIGNAL HEAD, SYMBOLIC PEDESTRIAN SIGNAL LAMP, ACCESSIBLE PEDESTRIAN SIGNALS, SIGNAL CONTROLLERS, SIGNAL CABINETS, BUS INTERFACE UNITS, BATTERY BACKUP UNITS. CHECK WEBSITE PERIODICALLY FOR CURRENT UPDATES.
32. THE CONTRACTOR IS RESPONSIBLE FOR THE SIGNAL CARRYING CAPABILITY AND PERFORMANCE OF THE CABLE. INSTALL EACH WIRE WITH A LIGHTNING PROTECTION DEVICE UNLESS OTHERWISE NOTED.
33. CONTRACTOR TO ADJUST SIGNAL HEAD ALIGNMENT, AS NEEDED, USING ARTICULATING SIGNAL BRACKET ASSEMBLIES WITH A MINIMUM OF THREE ADJUSTABLE AXES.
34. SEAL WITH WATERPROOF SEALANT EACH END OF THE COMMUNICATIONS CABLE THAT IS EXPOSED TO THE ELEMENTS DURING STORAGE OR AFTER INSTALLATION.
35. THE CONTRACTOR TO FURNISH AND INSTALL ALL EQUIPMENT CALLED FOR AND REQUIRED AS NEEDED FOR A FULLY OPERATIONAL TRAFFIC SIGNAL.
36. REMOVE THE EXISTING PAVEMENT MARKING AS DIRECTED. REMOVE THE PAVEMENT MARKING TO THE EXTENT THAT THEY ARE EITHER COMPLETELY REMOVED OR OBLITERATED TO THE SATISFACTION OF THE ENGINEER.
37. PLACE PAVEMENT MARKINGS AS SHOWN ON THE PLANS OR AS DIRECTED.
38. THE VENDOR'S REPRESENTATIVES OF THE RADAR EQUIPMENT SUPPLIED FOR THIS PROJECT MUST SUPERVISE THE INSTALLATION, SETUP AND TESTING OF THIS EQUIPMENT AND BE FACTORY CERTIFIED. THE REPRESENTATIVE MUST BE ON SITE DURING THIS TIME. ANY EQUIPMENT REQUIRED FOR SETUP AND OPERATION OF THE RADAR DEVICES MUST BE PROVIDED TO TXDOT UPON COMPLETION. THE VENDOR'S REPRESENTATIVE MUST PROVIDE TRAINING TO THE MUNICIPALITIES WHO WILL BE RESPONSIBLES FOR THE MAINTENANCE OF THE RADAR EQUIPMENT AFTER ACCEPTANCE OF THE PROJECT.
39. THE RADAR PRESENCE DETECTOR AND RADAR ADVANCE DETECTION DEVICES MUST BE COMPATIBLE WITH EACH OTHER AND FROM THE SAME MANUFACTURER.
40. RADAR PRESENCE DETECTION DEVICE MUST UTILIZE TRUE-PRESENCE DETECTION. SYSTEM USING LOCKING ALGORITHMS TO ATTEMPT PRESENCE DETECTION WILL NOT BE ACCEPTED.
41. RADAR ADVANCE DETECTION DEVICE MUST CONTINUOUSLY TRACK VEHICLE SPEED, DISTANCE, AND ESTIMATED TIME OF ARRIVAL.
42. COMMUNICATION AND POWER TO THE RADAR DEVICES SHALL BE VIA CONTINUOUS CABLE RUN OF UP TO 1000 FEET WITH THE USE OF REPEATERS.
43. THE LOCATIONS OF RADAR DEVICES SHOWN ON THIS PLAN ARE APPROXIMATE. THE EXACT PLACEMENT SHALL BE DETERMINED IN COORDINATION WITH THE MANUFACTURER'S REPRESENTATIVE. THE CONTRACTOR WILL VERIFY RADAR UNIT LOCATIONS WITH THE MANUFACTURER PRIOR TO INSTALLING DETECTION CABLE. FINAL PLACEMENT OF RADAR DEVICES IS SUBJECT TO APPROVAL BY THE ENGINEER.
44. THE LOCATION OF THE RADAR DETECTION ZONE IS APPROXIMATE. THE EXACT LOCATION WILL BE DETERMINED BY THE ENGINEER AND/OR DEPARTMENT'S TRAFFIC OPERATIONS SECTION.
45. EACH FIBER OPTIC CABLE RUN IN UNDERGROUND CONDUITS SHALL HAVE AN EXTRA LENGTH COILED AND LEFT IN EACH GROUND BOX. THE PROVIDED EXTRA LENGTH OF FIBER OPTIC CABLE SHALL BE PER THE PLANS OR AS DIRECTED BY THE ENGINEER IN THE FIELD.
46. LIMITS OF PAY FOR BORED CONDUITS SHALL NOT EXTEND MORE THAN FIVE FEET IN FRONT OF AND BEYOND THE ROADWAY OR DRIVEWAY THAT IS BEING BORED UNDER. WHEN MULTIPLE DRIVEWAYS EXIST, THE CONTRACTOR MAY BE ALLOWED TO BORE UNDER THE ENTIRE GROUP OF DRIVEWAYS PROVIDED THE DRIVEWAYS DO NOT EXCEED FORTY FOOT SPACING AS APPROVED BY THE ENGINEER IN THE FIELD. NO INCREASE FOR BORED CONDUITS WILL BE INCURRED FOR THIS WORK. CONDUIT BORED BETWEEN MULTIPLE DRIVEWAYS TO BE PAID FOR AS TRENCHED CONDUIT.
47. INSTALL A FIBER OPTIC PATCH PANEL AND CONNECTORS TO THE FIBER DROP CABLE IN EACH CONTROLLER CABINET. THESE FIBER OPTIC FUSION SPLICES SHALL BE CONSIDERED INCIDENTAL TO THE PERTINENT BID ITEMS IN THIS PROJECT.

STATE OF TEXAS
ANUSHA POKHREL
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PROFESSIONAL ENGINEER

4/29/2026

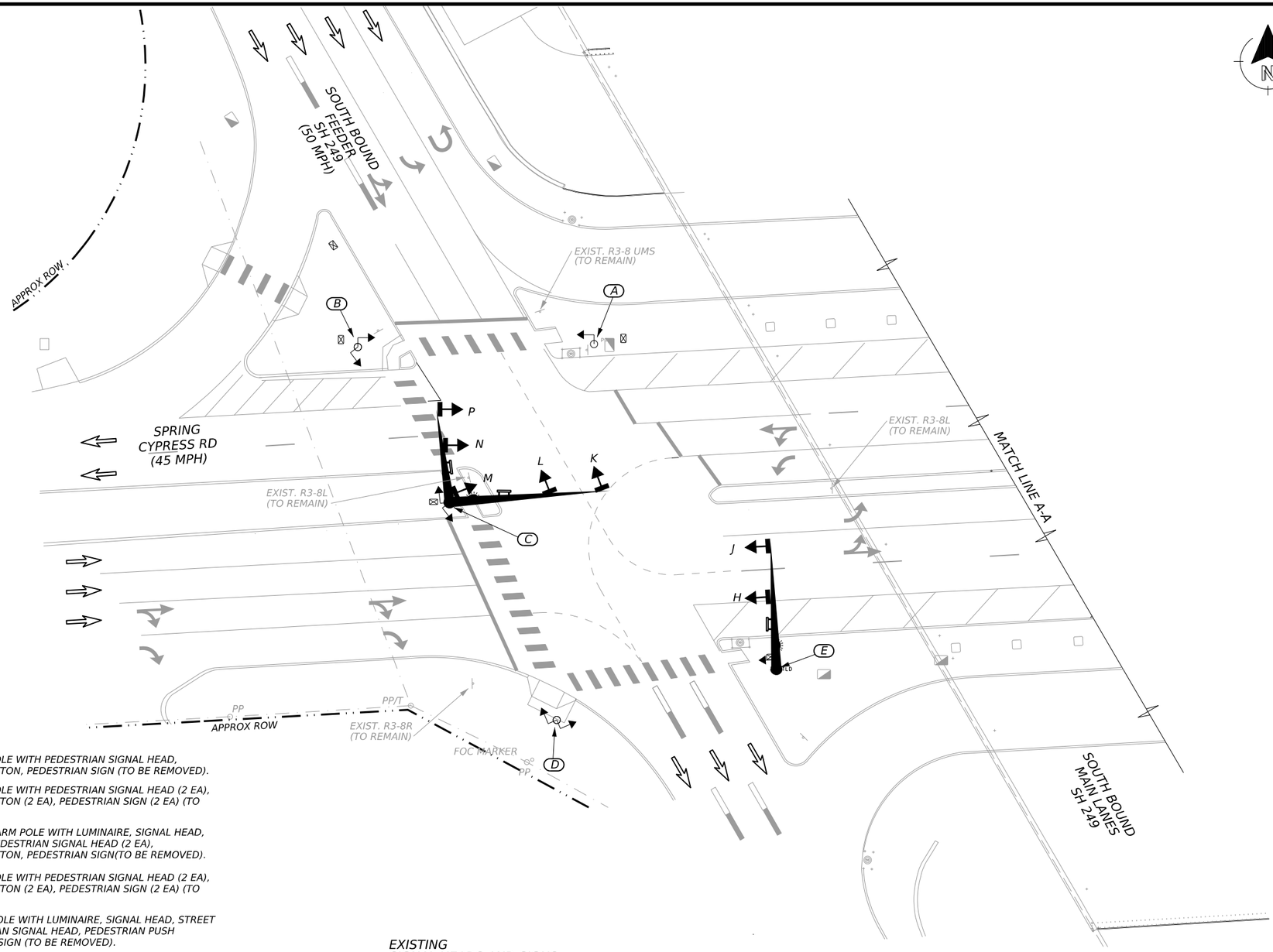
Texas Department of Transportation

TRAFFIC SIGNAL NOTES

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CONT	SECT	JOB	HIGHWAY
0720	03	153,ETC.	SH 249
DIST	COUNTY		SHEET NO.
HOU	HARRIS		23

DATE: 4/29/2026 9:27:13 AM
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LEGEND:

- TRAFFIC DIRECTION
- PP — POWER POLE
- PP/T — POWER POLE W/TRANSFORMER
- OP — OVERHEAD POWER LINE
- EXIST. SIGNAL HEAD
- EXIST. LUMINAIRE
- EXIST. SIGNAL CONTROLLER
- EXIST. GROUND BOX
- EXIST. PEDESTRIAN SIGNAL HEAD
- EXIST. MAST ARM POLE
- EXIST. PED PUSH BUTTON ON EXIST. TRAFFIC SIGNAL POLE
- EXIST. PEDESTRIAN POLE
- EXIST. ELECTRICAL SERVICE POLE
- EXIST. ITS GROUND BOX (TO REMAIN)
- FOC — EXIST. FIBER OPTIC CABLE

CALLOUTS:

(A) *EXISTING PEDESTAL POLE WITH PEDESTRIAN SIGNAL HEAD, PEDESTRIAN PUSH BUTTON, PEDESTRIAN SIGN (TO BE REMOVED).*

(B) *EXISTING PEDESTAL POLE WITH PEDESTRIAN SIGNAL HEAD (2 EA), PEDESTRIAN PUSH BUTTON (2 EA), PEDESTRIAN SIGN (2 EA) (TO BE REMOVED).*

(C) *EXISTING DUAL MAST ARM POLE WITH LUMINAIRE, SIGNAL HEAD, STREET NAME SIGN, PEDESTRIAN SIGNAL HEAD (2 EA), PEDESTRIAN PUSH BUTTON, PEDESTRIAN SIGN(TO BE REMOVED).*

(D) *EXISTING PEDESTAL POLE WITH PEDESTRIAN SIGNAL HEAD (2 EA), PEDESTRIAN PUSH BUTTON (2 EA), PEDESTRIAN SIGN (2 EA) (TO BE REMOVED).*

(E) *EXISTING MAST ARM POLE WITH LUMINAIRE, SIGNAL HEAD, STREET NAME SIGN, PEDESTRIAN SIGNAL HEAD, PEDESTRIAN PUSH BUTTON, PEDESTRIAN SIGN (TO BE REMOVED).*

(F) *EXISTING PEDESTAL POLE WITH PEDESTRIAN SIGNAL HEAD, PEDESTRIAN PUSH BUTTON, PEDESTRIAN SIGN (TO BE REMOVED).*

(G) *EXISTING PEDESTAL POLE WITH PEDESTRIAN SIGNAL HEAD (2 EA), PEDESTRIAN PUSH BUTTON (2 EA), PEDESTRIAN SIGN (2 EA) (TO BE REMOVED).*

(H) *EXISTING DUAL MAST ARM POLE WITH LUMINAIRE, SIGNAL HEAD, STREET NAME SIGN, PEDESTRIAN SIGNAL HEAD (2 EA), PEDESTRIAN PUSH BUTTON, PEDESTRIAN SIGN (TO BE REMOVED).*

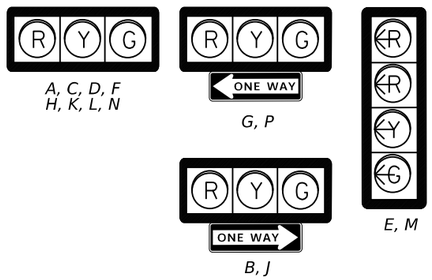
(J) *EXISTING PEDESTAL POLE WITH PEDESTRIAN SIGNAL HEAD (2 EA), PEDESTRIAN PUSH BUTTON (2 EA), PEDESTRIAN SIGN (2 EA) (TO BE REMOVED).*

(K) *EXISTING MAST ARM POLE WITH LUMINAIRE, SIGNAL HEAD, STREET NAME SIGN, PEDESTRIAN SIGNAL HEAD, PEDESTRIAN PUSH BUTTON, PEDESTRIAN SIGN (TO BE REMOVED).*

(L) *EXISTING SIGNAL CONTROLLER (TO BE REMOVED).*

(M) *EXISTING ELECTRICAL SERVICE POLE (VERIFY AND REMOVED IF NOT IN SERVICE).*

EXISTING SIGNAL HEADS AND SIGNS (TO BE REMOVED)



0 20 40 FT

SCALE: 1" = 40'

STATE OF TEXAS
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4/29/2026

Texas Department of Transportation

SH 249 AT SPRING CYPRESS RD

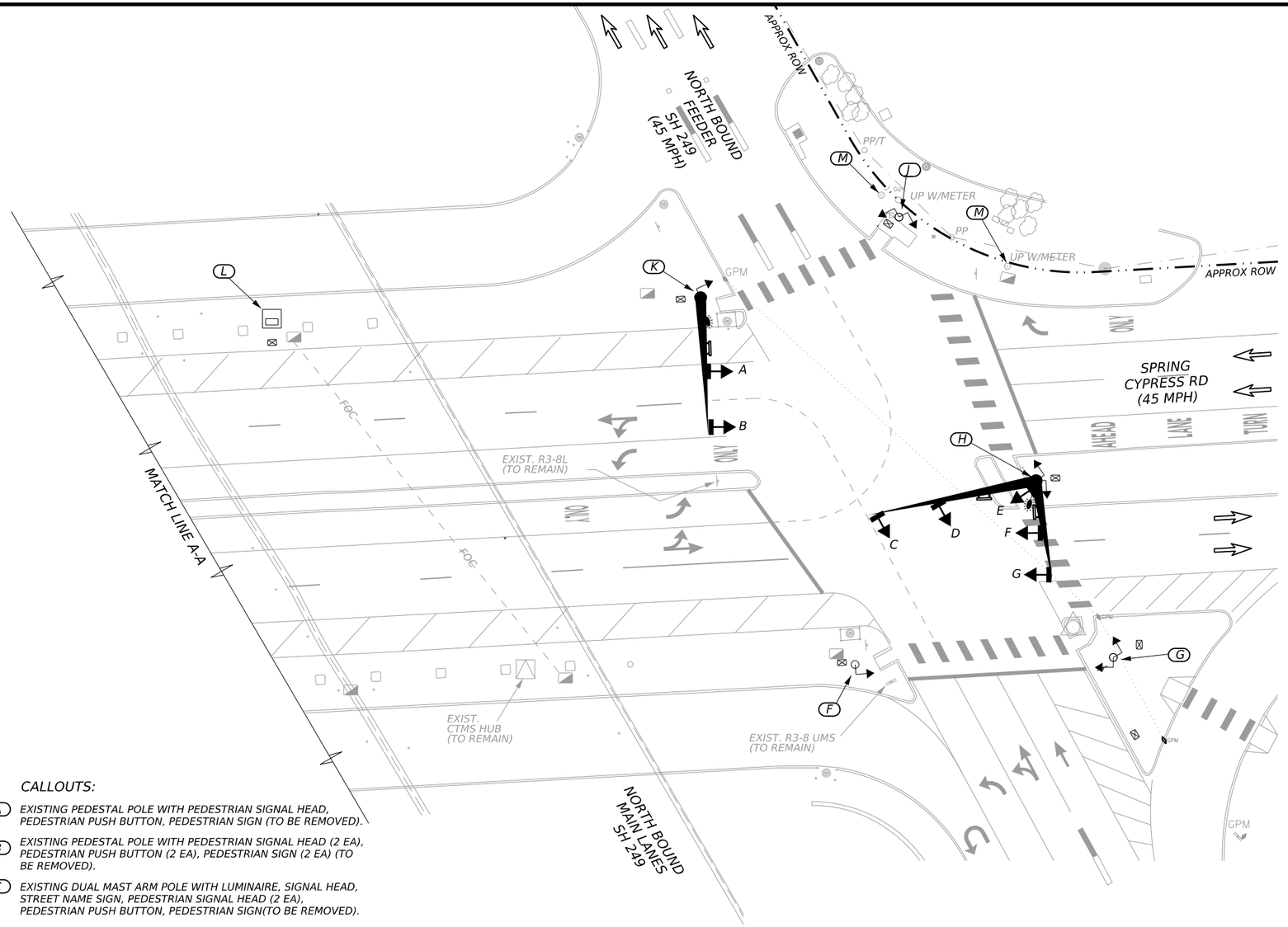
TRAFFIC SIGNAL EXISTING PLAN

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SHEET 1 OF 2

CONT	SECT	JOB	HIGHWAY
0720	03	153,ETC.	SH 249
DIST	COUNTY		SHEET NO.
HOU	HARRIS		24

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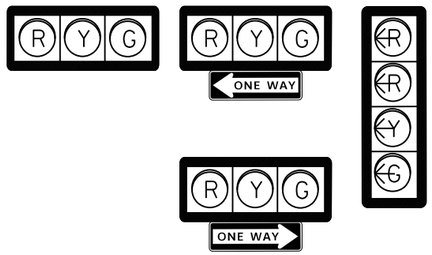
LEGEND:

- TRAFFIC DIRECTION
- PP — POWER POLE
- PP/T — POWER POLE W/TRANSFORMER
- OP — OVERHEAD POWER LINE
- EXIST. SIGNAL HEAD
- EXIST. LUMINAIRE
- EXIST. SIGNAL CONTROLLER
- EXIST. GROUND BOX
- EXIST. PEDESTRIAN SIGNAL HEAD
- EXIST. MAST ARM POLE
- EXIST. PED PUSH BUTTON ON EXIST. TRAFFIC SIGNAL POLE
- EXIST. PEDESTRIAN POLE
- EXIST. ELECTRICAL SERVICE POLE
- EXIST. ITS GROUND BOX (TO REMAIN)
- FOC — EXIST. FIBER OPTIC CABLE

CALLOUTS:

- (A) EXISTING PEDESTAL POLE WITH PEDESTRIAN SIGNAL HEAD, PEDESTRIAN PUSH BUTTON, PEDESTRIAN SIGN (TO BE REMOVED).
- (B) EXISTING PEDESTAL POLE WITH PEDESTRIAN SIGNAL HEAD (2 EA), PEDESTRIAN PUSH BUTTON (2 EA), PEDESTRIAN SIGN (2 EA) (TO BE REMOVED).
- (C) EXISTING DUAL MAST ARM POLE WITH LUMINAIRE, SIGNAL HEAD, STREET NAME SIGN, PEDESTRIAN SIGNAL HEAD (2 EA), PEDESTRIAN PUSH BUTTON, PEDESTRIAN SIGN(TO BE REMOVED).
- (D) EXISTING PEDESTAL POLE WITH PEDESTRIAN SIGNAL HEAD (2 EA), PEDESTRIAN PUSH BUTTON (2 EA), PEDESTRIAN SIGN (2 EA) (TO BE REMOVED).
- (E) EXISTING MAST ARM POLE WITH LUMINAIRE, SIGNAL HEAD, STREET NAME SIGN, PEDESTRIAN SIGNAL HEAD, PEDESTRIAN PUSH BUTTON, PEDESTRIAN SIGN (TO BE REMOVED).
- (F) EXISTING PEDESTAL POLE WITH PEDESTRIAN SIGNAL HEAD, PEDESTRIAN PUSH BUTTON, PEDESTRIAN SIGN (TO BE REMOVED).
- (G) EXISTING PEDESTAL POLE WITH PEDESTRIAN SIGNAL HEAD (2 EA), PEDESTRIAN PUSH BUTTON (2 EA), PEDESTRIAN SIGN (2 EA) (TO BE REMOVED).
- (H) EXISTING DUAL MAST ARM POLE WITH LUMINAIRE, SIGNAL HEAD, STREET NAME SIGN, PEDESTRIAN SIGNAL HEAD (2 EA), PEDESTRIAN PUSH BUTTON, PEDESTRIAN SIGN (TO BE REMOVED).
- (I) EXISTING PEDESTAL POLE WITH PEDESTRIAN SIGNAL HEAD (2 EA), PEDESTRIAN PUSH BUTTON (2 EA), PEDESTRIAN SIGN (2 EA) (TO BE REMOVED).
- (K) EXISTING MAST ARM POLE WITH LUMINAIRE, SIGNAL HEAD, STREET NAME SIGN, PEDESTRIAN SIGNAL HEAD, PEDESTRIAN PUSH BUTTON, PEDESTRIAN SIGN (TO BE REMOVED).
- (L) EXISTING SIGNAL CONTROLLER (TO BE REMOVED).
- (M) EXISTING ELECTRICAL SERVICE POLE (VERIFY AND REMOVED IF NOT IN SERVICE).

EXISTING SIGNAL HEADS AND SIGNS (TO BE REMOVED)



0 20 40 FT

SCALE: 1" = 40'

STATE OF TEXAS
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4/29/2026

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SHEET 2 OF 2

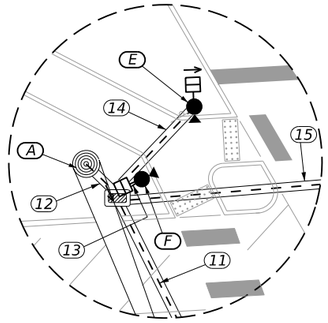
Texas Department of Transportation

SH 249
AT SPRING CYPRESS RD

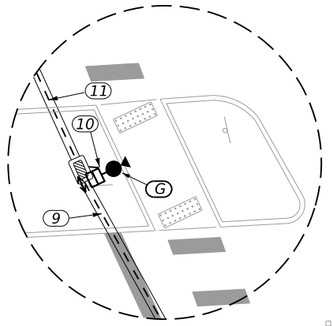
TRAFFIC SIGNAL EXISTING PLAN

CONT	SECT	JOB	HIGHWAY
0720	03	153,ETC.	SH 249
DIST	COUNTY		SHEET NO.
HOU	HARRIS		25

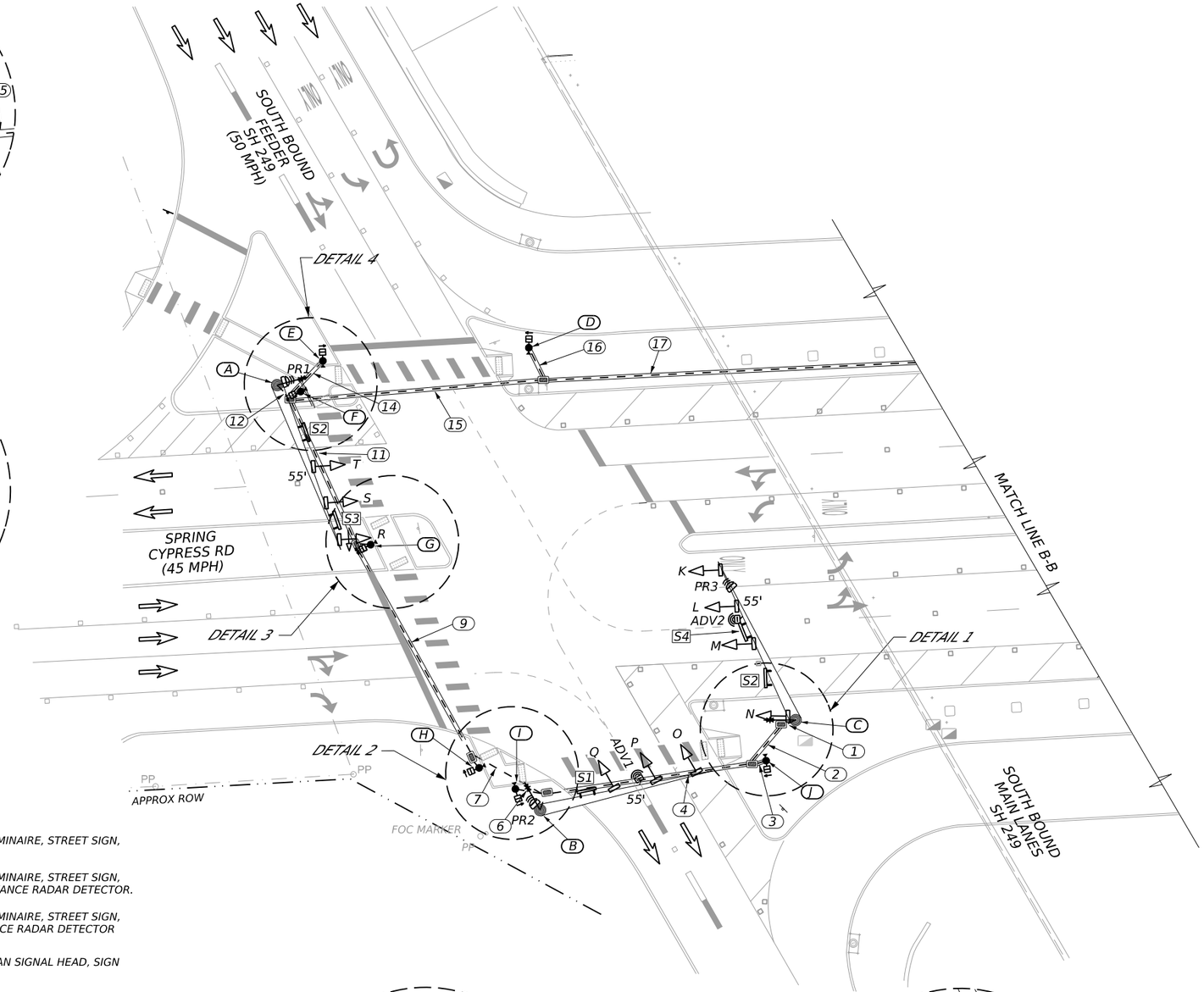
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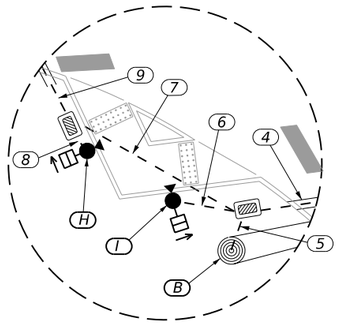


DETAIL 4
(N.T.S.)

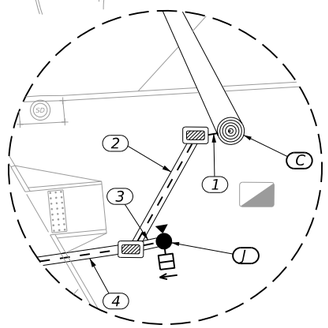


DETAIL 3
(N.T.S.)





DETAIL 2
(N.T.S.)



DETAIL 1
(N.T.S.)

CALLOUTS:

- (A) PROP. 55 FT. MAST ARM POLE WITH LUMINAIRE, STREET SIGN, AND PRESENCE RADAR DETECTOR.
- (B) PROP. 55 FT. MAST ARM POLE WITH LUMINAIRE, STREET SIGN, PRESENCE RADAR DETECTOR AND ADVANCE RADAR DETECTOR.
- (C) PROP. 55 FT. MAST ARM POLE WITH LUMINAIRE, STREET SIGN, POLE-MOUNTED SIGNAL HEAD, PRESENCE RADAR DETECTOR AND ADVANCE RADAR DETECTOR.
- (D, E, F, H, I, J) PROP. PEDESTAL POLE WITH PEDESTRIAN SIGNAL HEAD, SIGN AND PUSH BUTTON (APS UNIT).
- (G) PROP. PEDESTAL POLE WITH PEDESTRIAN SIGNAL HEADS (2 EA), SIGN AND PUSH BUTTON (APS UNIT).

NOTES:

THE CONTRACTOR SHALL LOCATE ALL UNDERGROUND AND ABOVE GROUND UTILITIES PRIOR TO COMMENCING WORK. THE CONTRACTOR SHALL CONTACT PUBLIC AND PRIVATE UTILITIES AT LEAST 72 HOURS PRIOR TO ANY WORK. TXDOT IS NOT A MEMBER OF 811, THE CONTRACTOR SHALL CONTACT TXDOT FIVE (5) BUSINESS DAYS TO LOCATED TXDOT OWNED EXISTING TXDOT COMMUNICATIONS, ILLUMINATION, AND TRAFFIC SIGNAL CABLING. TXDOT HOUSTON DISTRICT TRAFFIC OPERATIONS OFFICE CAN BE REACHED AT: HOU-LOCATEREQUEST@TXDOT.GOV

THE CONTRACTOR SHALL BE FULLY RESPONSIBLE FOR ANY DAMAGES CAUSED BY CONTRACTOR'S FAILURE TO LOCATE AND PRESERVE THESE UTILITIES WHETHER UNDERGROUND OR ABOVE GROUND. UTILITIES ON THE PLANS ARE SHOWN IN APPROXIMATE LOCATIONS.

LEGEND:

- TRAFFIC DIRECTION
- PP — POWER POLE
- PP/T — POWER POLE W/TRANSFORMER
- OP — OVERHEAD POWER LINE
- PROP. MAST ARM POLE
- PROP. LUMINAIRE
- PROP. SIGNAL HEAD
- PROP. TURN LANE SIGNAL HEAD
- ADV1 — PROP. ADVANCE RADAR DETECTOR
- PRE1 — PROP. PRESENCE RADAR DETECTOR
- PROP. PEDESTRIAN SIGNAL HEAD
- PROP. PEDESTRIAN POLE WITH PUSH BUTTON
- PROP. CONDUIT (TRENCH)
- PROP. CONDUIT (BORED)
- PROP. TRAFFIC SIGNAL CONTROLLER
- PROP. GROUND BOX
- PROP. STREET NAME SIGN
- PROP. ELECTRICAL SERVICE POLE
- EXIST. SIGNAL CONTROLLER
- EXIST. ITS GROUND BOX (TO REMAIN)
- FOC — EXIST. FIBER OPTIC CABLE

0 20 40 FT
SCALE: 1" = 40'

STATE OF TEXAS
ANUSHA POKHREL
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LICENSED
PROFESSIONAL ENGINEER

4/29/2026

Texas Department of Transportation

SH 249
AT SPRING CYPRESS RD

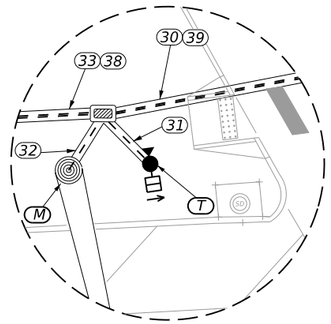
TRAFFIC SIGNAL
PROPOSED PLAN

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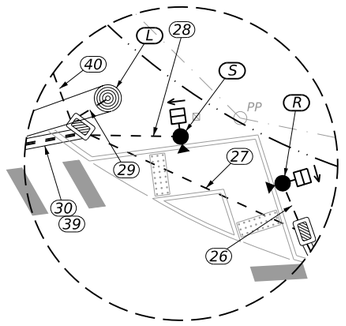
SHEET 1 OF 4

CONT	SECT	JOB	HIGHWAY
0720	03	153,ETC.	SH 249
DIST	COUNTY		SHEET NO.
HOU	HARRIS		26

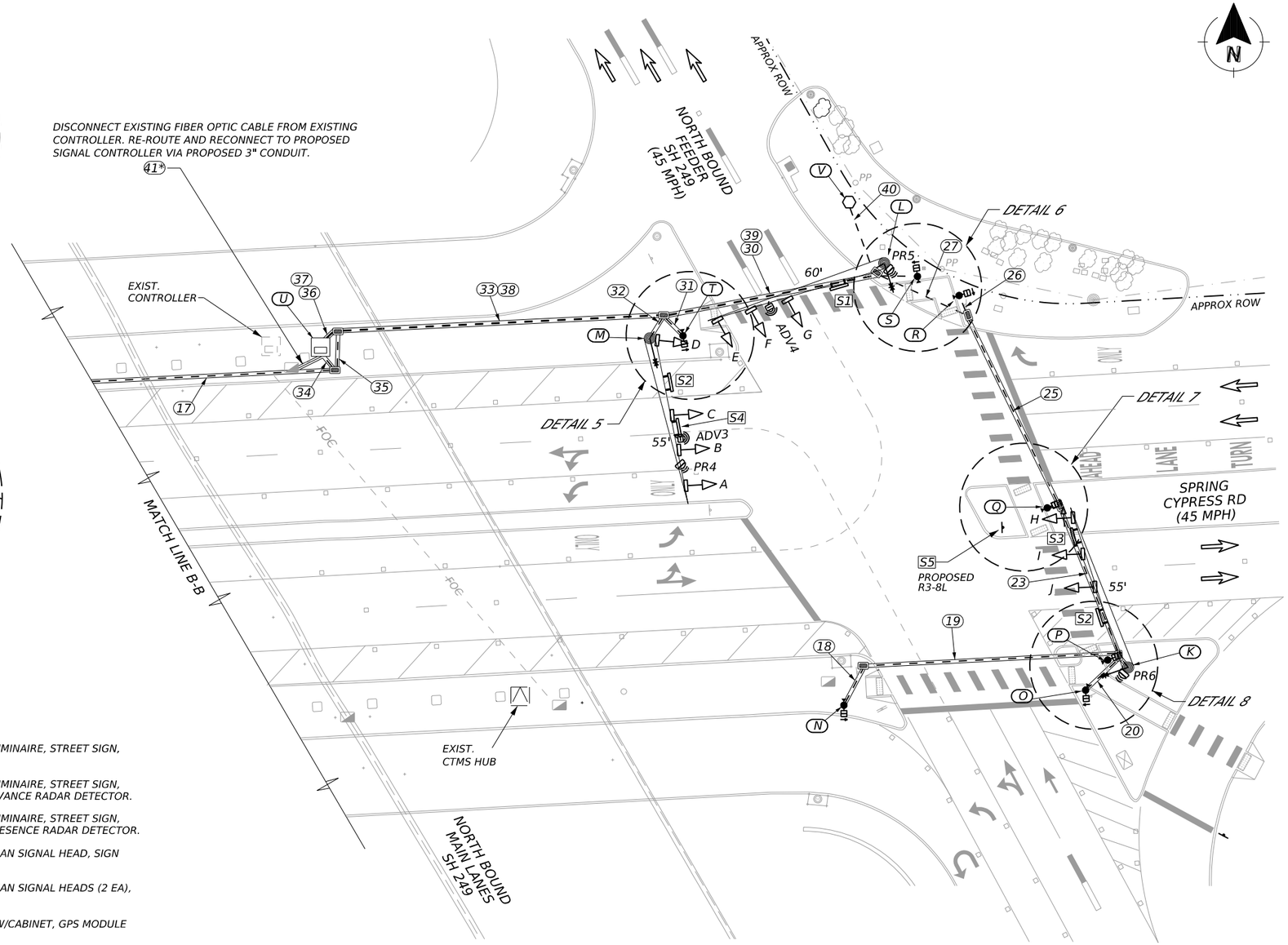
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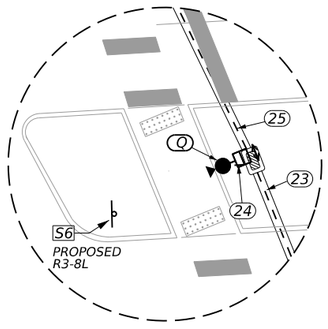


DETAIL 5
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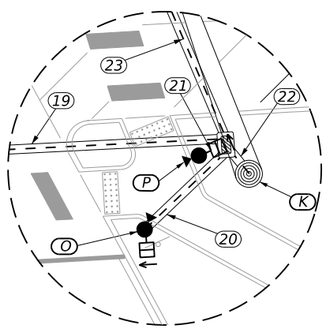


DETAIL 6
(N.T.S.)





DETAIL 7
(N.T.S.)



DETAIL 8
(N.T.S.)

CALLOUTS:

- (K) PROP. 55 FT. MAST ARM POLE WITH LUMINAIRE, STREET SIGN, AND PRESENCE RADAR DETECTOR.
- (L) PROP. 60 FT. MAST ARM POLE WITH LUMINAIRE, STREET SIGN, PRESENCE RADAR DETECTOR AND ADVANCE RADAR DETECTOR.
- (M) PROP. 55 FT. MAST ARM POLE WITH LUMINAIRE, STREET SIGN, POLE-MOUNTED SIGNAL HEAD AND PRESENCE RADAR DETECTOR.
- (N, O, P, R, S, T) PROP. PEDESTAL POLE WITH PEDESTRIAN SIGNAL HEAD, SIGN AND PUSH BUTTON (APS UNIT).
- (Q) PROP. PEDESTAL POLE WITH PEDESTRIAN SIGNAL HEADS (2 EA), SIGN AND PUSH BUTTON (APS UNIT).
- (U) PROP. FULL-ACTUATED CONTROLLER W/CABINET, GPS MODULE AND BATTERY BACK-UP (BBU)
- (V) PROP. ELECTRICAL SERVICE POLE TYPE D WITH METER, SERVICE (120/240 VOLT) AND SERVICE DISCONNECT

NOTES:

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LEGEND:

- TRAFFIC DIRECTION
- PP — POWER POLE
- PP/T — POWER POLE W/TRANSFORMER
- OP — OVERHEAD POWER LINE
- PROP. MAST ARM POLE
- PROP. LUMINAIRE
- PROP. SIGNAL HEAD
- PROP. TURN LANE SIGNAL HEAD
- ADV1 — PROP. ADVANCE RADAR DETECTOR
- PRE1 — PROP. PRESENCE RADAR DETECTOR
- PROP. PEDESTRIAN SIGNAL HEAD
- PROP. PEDESTRIAN POLE WITH PUSH BUTTON
- PROP. CONDUIT (TRENCH)
- PROP. CONDUIT (BORED)
- PROP. TRAFFIC SIGNAL CONTROLLER
- PROP. GROUND BOX
- PROP. STREET NAME SIGN
- PROP. ELECTRICAL SERVICE POLE
- EXIST. SIGNAL CONTROLLER
- EXIST. ITS GROUND BOX (TO REMAIN)
- FOC — EXIST. FIBER OPTIC CABLE

0 20 40 FT
SCALE: 1" = 40'

STATE OF TEXAS
ANUSHA POKHREL
151136
LICENSED
PROFESSIONAL ENGINEER

4/29/2026

Texas Department of Transportation

SH 249
AT SPRING CYPRESS RD

TRAFFIC SIGNAL PROPOSED PLAN

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SHEET 2 OF 4

CONT	SECT	JOB	HIGHWAY
0720	03	153,ETC.	SH 249
DIST	COUNTY		SHEET NO.
HOU	HARRIS		27

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PROPOSED PEDESTRIAN SIGNAL HEAD AND PUSH BUTTON ORIENTATION:



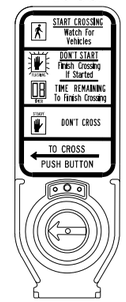
PROPOSED PEDESTRIAN SIGNAL HEAD (COUNTDOWN TYPE):



P1, P2, P3, P4a, P4b, P5, P6, P7
P8, P9, P10, P11a, P11b, P12, P13, P14

PROPOSED PEDESTRIAN SIGNS AND PUSH BUTTONS:

R10-3eL



PB1, PB3, PB6, PB8, PB10, PB13, PB14

R10-3eD



PB4, PB11

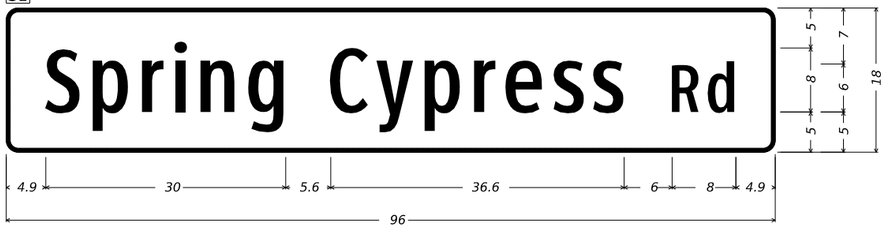
R10-3eR



PB2, PB5, PB7, PB9, PB12

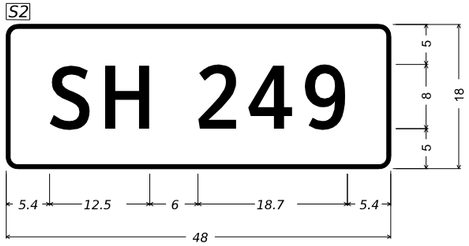
PROPOSED STREET SIGN:

S1



1.5" Radius, 0.5" Border, White on Green;
"Spring Cypress", ClearviewHwy-2-W 85% spacing; "Rd", ClearviewHwy-2-W 85% spacing;

S2



1.5" Radius, 0.5" Border, White on Green;
"SH", ClearviewHwy-3-W; "249", ClearviewHwy-3-W;

S3



R6-1L 36x12;

S4



R6-1R 36x12;

PROPOSED RADAR DETECTIONS SCHEDULE:

ADV1	DESIGNATED FOR SOUTHBOUND APPROACHING VEHICLES (SH 249)(FEEDER RD)
ADV2	DESIGNATED FOR EASTBOUND APPROACHING VEHICLES (SPRING CYPRESS RD)
ADV3	DESIGNATED FOR WESTBOUND APPROACHING VEHICLES (SPRING CYPRESS RD)
ADV4	DESIGNATED FOR NORTHBOUND APPROACHING VEHICLES (SH 249)(FEEDER RD)
PR1	DESIGNATED FOR PRESENCE OF SOUTHBOUND VEHICLES (SH 249)(FEEDER RD)
PR2	DESIGNATED FOR PRESENCE OF EASTBOUND VEHICLES (SPRING CYPRESS RD)
PR3	DESIGNATED FOR PRESENCE OF WESTBOUND VEHICLES (SPRING CYPRESS RD)
PR4	DESIGNATED FOR PRESENCE OF EASTBOUND VEHICLES (SPRING CYPRESS RD)
PR5	DESIGNATED FOR PRESENCE OF WESTBOUND VEHICLES (SPRING CYPRESS RD)
PR6	DESIGNATED FOR PRESENCE OF NORTHBOUND VEHICLES (SH 249)(FEEDER RD)

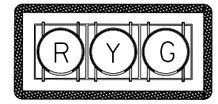
PROPOSED SMALL SIGN:

S5

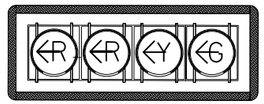


R3-8L_30x30;

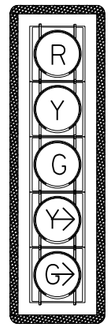
PROPOSED SIGNAL HEADS:



A, B, C, E, F, G, I, J,
K, L, M, O, P, Q, S, T



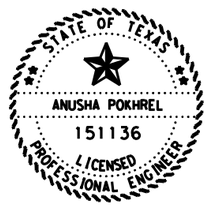
H, R



D, N

ELECTRICAL SERVICE DATA:

ELECTRICAL SERVICE NAME	CALLOUT	ELECTRICAL SERVICE DESCRIPTION SEE ED (5, 6, 7 & 8)-25	SERVICE CONDUIT SIZE (RMC)	SERVICE CONDUCTORS NO./SIZE	SAFETY SWITCH AMPS	MAIN CKT. BRK. POLE/AMP	TWO-POLE CONTACTOR AMPS	PANELBD./ LOAD CENTER AMP RATING (MIN)	CIRCUIT NO.	BRANCH CKT. BRK. POLE/AMPS	BRANCH CIRCUIT AMPS	kVA LOAD
SH 249 AT SPRING CYPRESS RD	V	ELEC SERV TY D (120/240)060(NS)SS(E)SP(O)	1-1/4"	3/#6	N/A	2P/60		100	TRF. SIG	1P/50	40	6.7
							30		LIGHTING	2P/20	6	



4/29/2026

Texas Department of Transportation

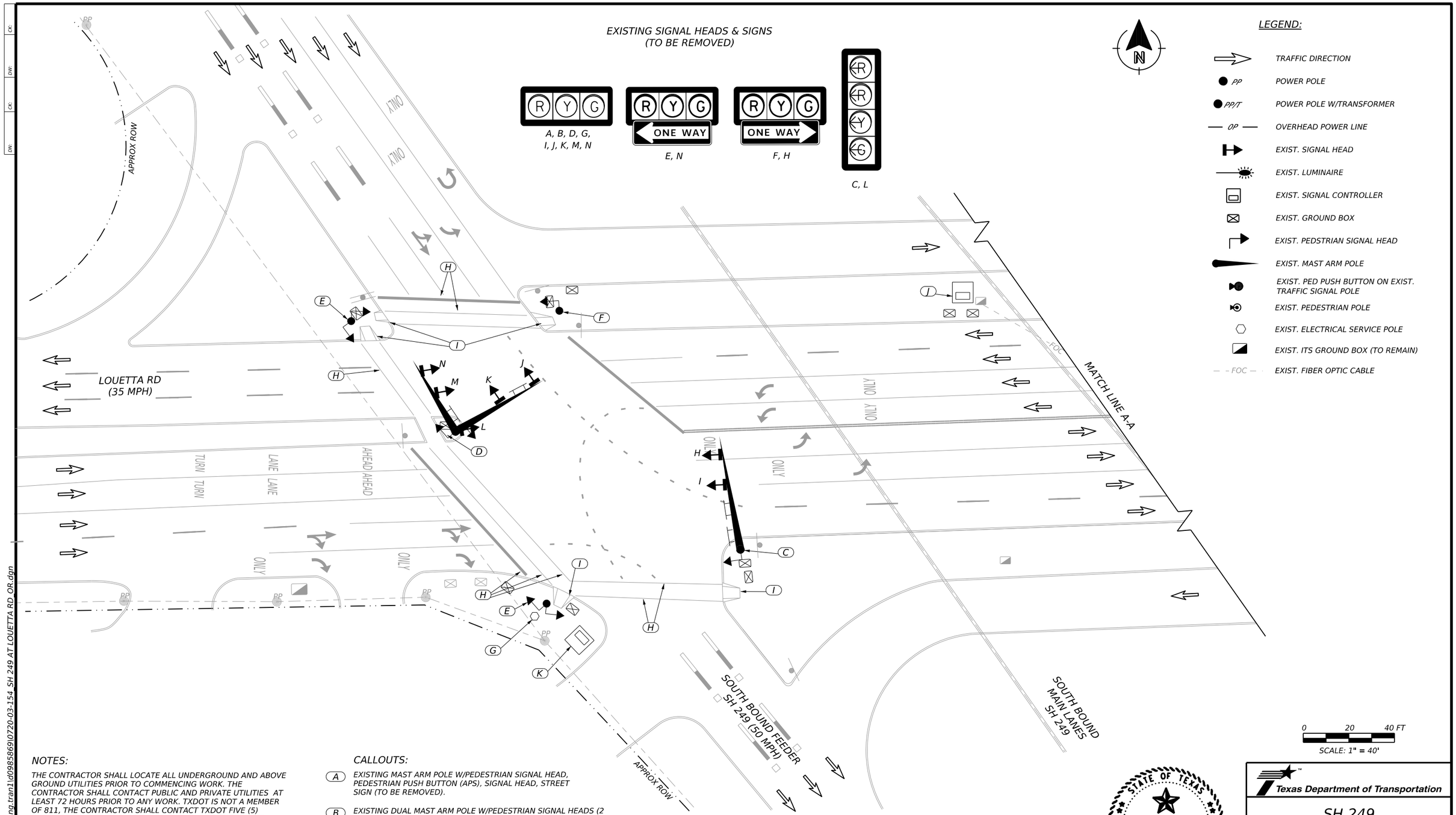
SH 249
AT SPRING CYPRESS RD

TRAFFIC SIGNAL
PROPOSED PLAN

©TxDOT 2026

CONT	SECT	JOB	HIGHWAY
0720	03	153,ETC.	SH 249
DIST	COUNTY		SHEET NO.
HOU	HARRIS		28

DATE: 4/29/2026 9:27:38 AM
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DATE: 4/29/2026 9:27:50 AM
FILE: c:\txdot\pw_online\txdot3\hoang.tran1\d0985869\0720-03-154_SH 249 AT LOUETTA RD_OR.dgn

NOTES:

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CALLOUTS:

(A) EXISTING MAST ARM POLE W/PEDESTRIAN SIGNAL HEAD, PEDESTRIAN PUSH BUTTON (APS), SIGNAL HEAD, STREET SIGN (TO BE REMOVED).

(B) EXISTING DUAL MAST ARM POLE W/PEDESTRIAN SIGNAL HEADS (2 EA.), PEDESTRIAN PUSH BOTTONS (2 EA.)(APS), PEDESTRIAN SIGNS (R10-3e), SIGNAL HEAD, STREET SIGN (TO BE REMOVED).

(C) EXISTING MAST ARM POLE W/PEDESTRIAN SIGNAL HEAD PEDESTRIAN PUSH BUTTON (APS), SIGNAL HEAD, STREET SIGN (TO BE REMOVED).

(D) EXISTING DUAL MAST ARM POLE W/PEDESTRIAN SIGNAL HEADS (2 EA.), PEDESTRIAN PUSH BOTTONS (2 EA.)(APS), PEDESTRIAN SIGNS (R10-3e), SIGNAL HEAD, STREET SIGN (TO BE REMOVED).

(E) EXISTING PEDESTAL POLE WITH PEDESTRIAN SIGNAL HEAD (2 EA), PEDESTRIAN PUSH BUTTON (2 EA), PEDESTRIAN SIGN (2 EA) (TO BE REMOVED).

(F) EXISTING PEDESTAL POLE W/PEDESTRIAN SIGNAL HEAD AND PUSH BOTTON (TO BE REMOVED).

(G) EXISTING ELECTRICAL SERVICE POLE (VERIFY AND REMOVED IF NOT IN SERVICE).

(H) EXISTING PAVEMENT MARKING (TO BE REMOVED.)

(I) EXISTING RAMP (TO BE REMOVED).

(J) EXISTING SIGNAL CONTROLLER (TO BE REMOVED).

(K) ABANDONED SIGNAL CONTROLLER (VERIFY AND REMOVED IF NOT IN SERVICE).

STATE OF TEXAS
ANUSHA POKHREL
151136
REGISTERED
PROFESSIONAL ENGINEER

4/29/2026

Texas Department of Transportation

SH 249
AT LOUETTA RD

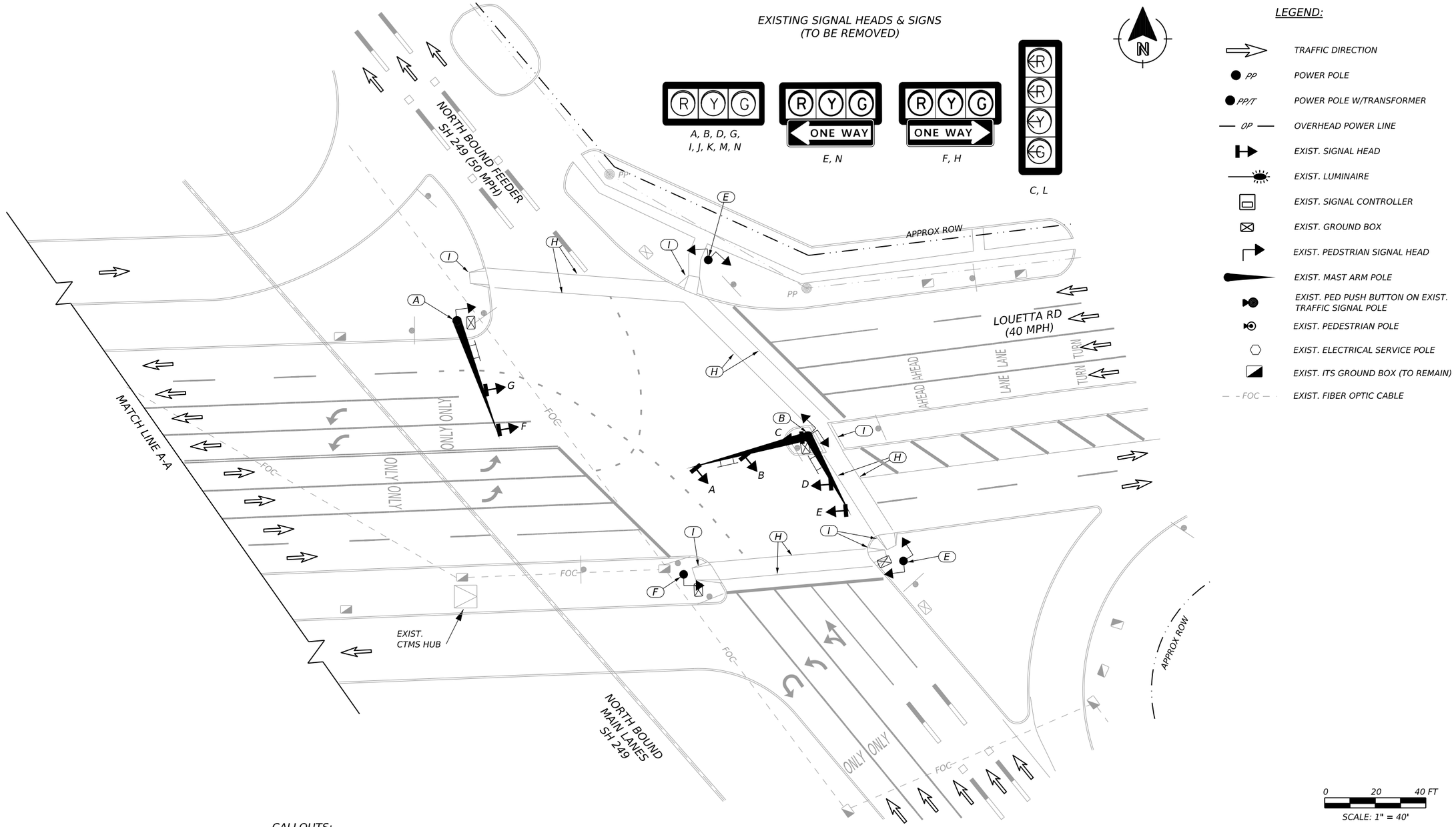
TRAFFIC SIGNAL EXISTING PLAN

©TxDOT 2026

SHEET 1 OF 2

CONT	SECT	JOB	HIGHWAY
0720	03	153,ETC.	SH 249

DIST	COUNTY	SHEET NO.
HOU	HARRIS	30



0 20 40 FT

SCALE: 1" = 40'

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CALLOUTS:

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(B) EXISTING DUAL MAST ARM POLE W/PEDESTRIAN SIGNAL HEADS (2 EA.), PEDESTRIAN PUSH BOTTONS (2 EA.)(APS), PEDESTRIAN SIGNS (R10-3e), SIGNAL HEAD, STREET SIGN (TO BE REMOVED).

(C) EXISTING MAST ARM POLE W/PEDESTRIAN SIGNAL HEAD PEDESTRIAN PUSH BUTTON (APS), SIGNAL HEAD, STREET SIGN (TO BE REMOVED).

(D) EXISTING DUAL MAST ARM POLE W/PEDESTRIAN SIGNAL HEADS (2 EA.), PEDESTRIAN PUSH BOTTONS (2 EA.)(APS), PEDESTRIAN SIGNS (R10-3e), SIGNAL HEAD, STREET SIGN (TO BE REMOVED).

(E) EXISTING PEDESTAL POLE WITH PEDESTRIAN SIGNAL HEAD (2 EA), PEDESTRIAN PUSH BUTTON (2 EA), PEDESTRIAN SIGN (2 EA) (TO BE REMOVED).

(F) EXISTING PEDESTAL POLE W/PEDESTRIAN SIGNAL HEAD AND PUSH BOTTON (TO BE REMOVED).

(G) EXISTING ELECTRICAL SERVICE POLE (VERIFY AND REMOVED IF NOT IN SERVICE).

(H) EXISTING PAVEMENT MARKING (TO BE REMOVED.)

(I) EXISTING RAMP (TO BE REMOVED).

(J) EXISTING SIGNAL CONTROLLER (TO BE REMOVED).

(K) ABANDONED SIGNAL CONTROLLER (VERIFY AND REMOVED IF NOT IN SERVICE).

STATE OF TEXAS
ANUSHA POKHREL
151136
REGISTERED
PROFESSIONAL ENGINEER

4/29/2026

Texas Department of Transportation

SH 249
AT LOUETTA RD

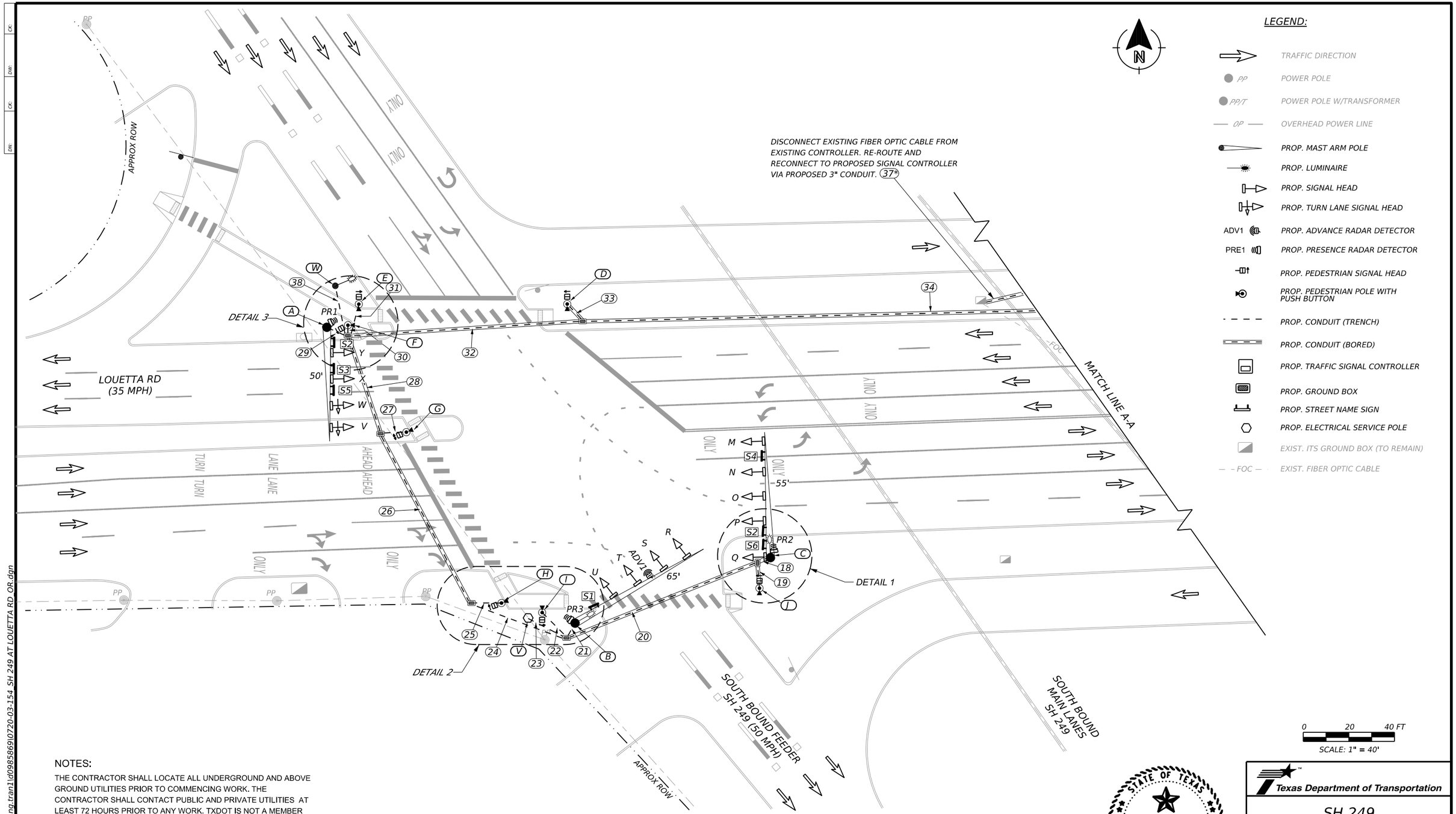
TRAFFIC SIGNAL EXISTING PLAN

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SHEET 2 OF 2

CONT	SECT	JOB	HIGHWAY
0720	03	153,ETC.	SH 249
DIST	COUNTY		SHEET NO.
HOU	HARRIS		31

DATE: 4/29/2026 9:27:50 AM
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STATE OF TEXAS
ANUSHA POKHREL
151136
REGISTERED
PROFESSIONAL ENGINEER

4/29/2026

Texas Department of Transportation

SH 249
AT LOUETTA RD

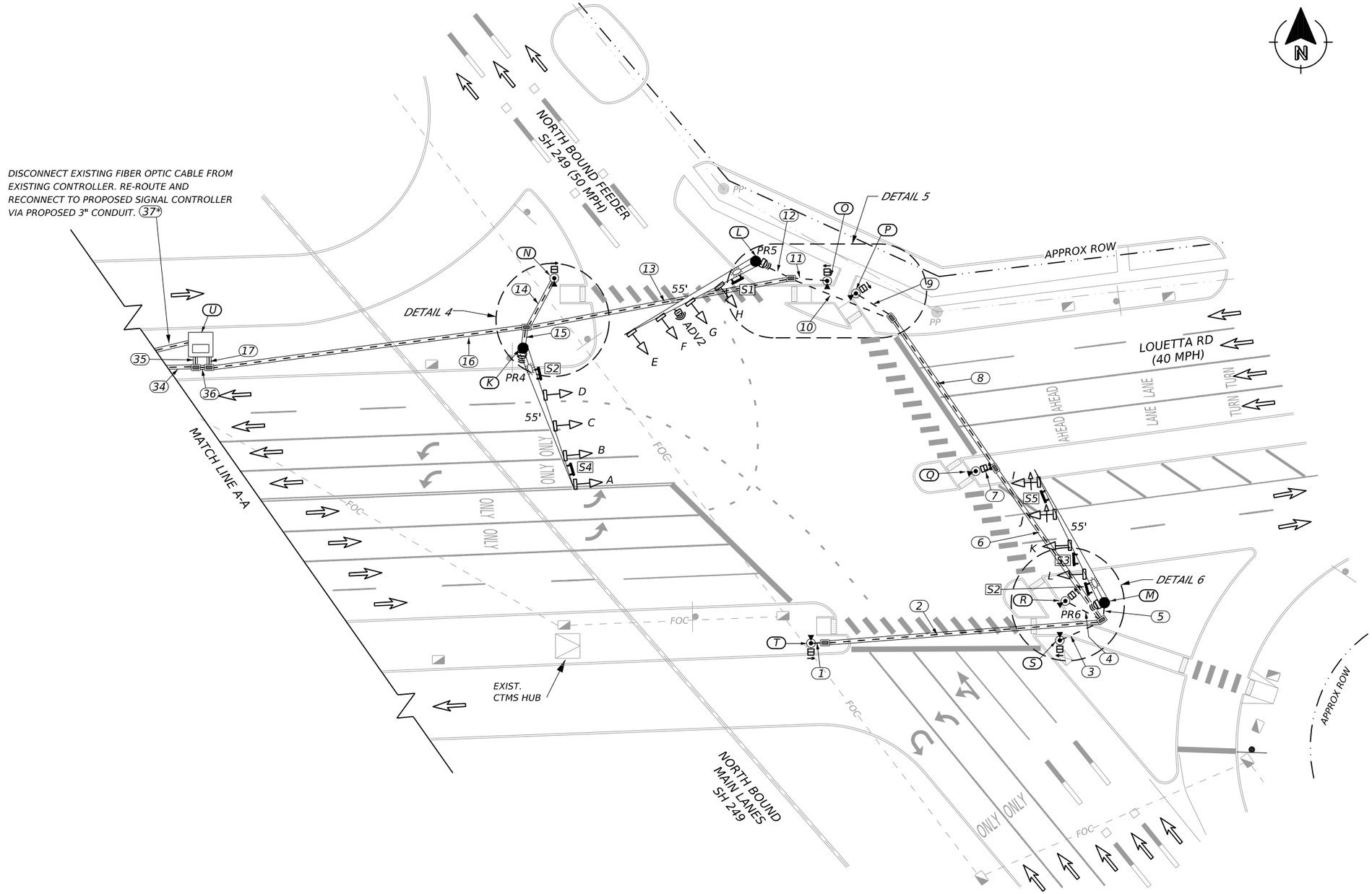
TRAFFIC SIGNAL PROPOSED PLAN

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SHEET 1 OF 6

CONT	SECT	JOB	HIGHWAY
0720	03	153,ETC.	SH 249
DIST	COUNTY		SHEET NO.
HOU	HARRIS		32

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LEGEND:

- TRAFFIC DIRECTION
- PP — POWER POLE
- PP/T — POWER POLE W/TRANSFORMER
- OP — OVERHEAD POWER LINE
- PROP. MAST ARM POLE
- PROP. LUMINAIRE
- PROP. SIGNAL HEAD
- PROP. TURN LANE SIGNAL HEAD
- ADV1 — PROP. ADVANCE RADAR DETECTOR
- PRE1 — PROP. PRESENCE RADAR DETECTOR
- PROP. PEDESTRIAN SIGNAL HEAD
- PROP. PEDESTRIAN POLE WITH PUSH BUTTON
- PROP. CONDUIT (TRENCH)
- PROP. CONDUIT (BORED)
- PROP. TRAFFIC SIGNAL CONTROLLER
- PROP. GROUND BOX
- PROP. STREET NAME SIGN
- PROP. ELECTRICAL SERVICE POLE
- EXIST. ITS GROUND BOX (TO REMAIN)
- – FOC — EXIST. FIBER OPTIC CABLE

0 20 40 FT

SCALE: 1" = 40'

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STATE OF TEXAS
ANUSHA POKHREL
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PROFESSIONAL ENGINEER

4/29/2026

Texas Department of Transportation

SH 249
AT LOUETTA RD

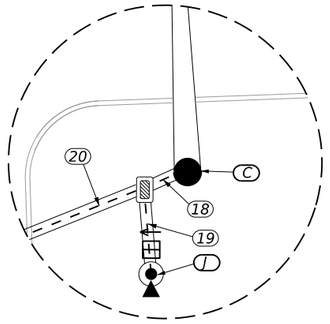
TRAFFIC SIGNAL PROPOSED PLAN

©TxDOT 2026

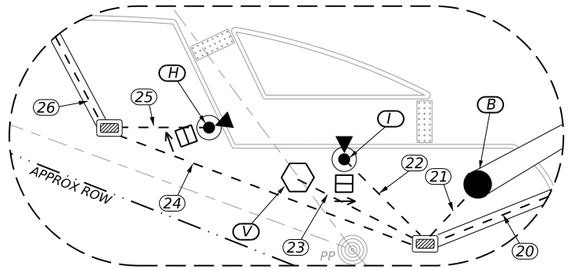
SHEET 2 OF 6

CONT	SECT	JOB	HIGHWAY
0720	03	153,ETC.	SH 249
DIST	COUNTY		SHEET NO.
HOU	HARRIS		33

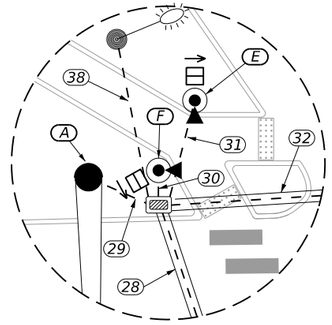
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DETAIL 1
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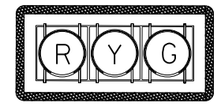


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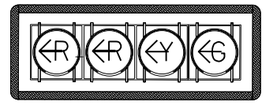


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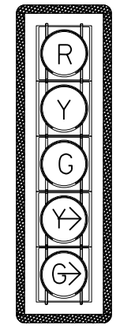
PROPOSED SIGNAL HEADS:



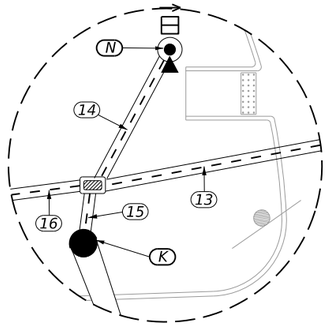
A, B, C, D, E, F, G, H, K, L,
M, N, O, P, R, S, T, U, X, Y



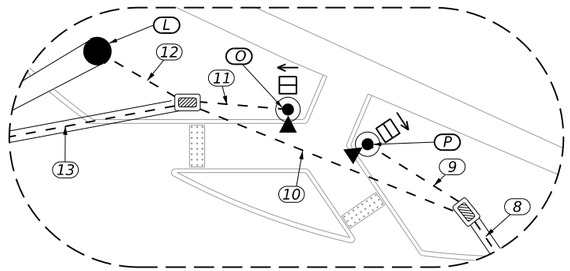
I, J, V, W



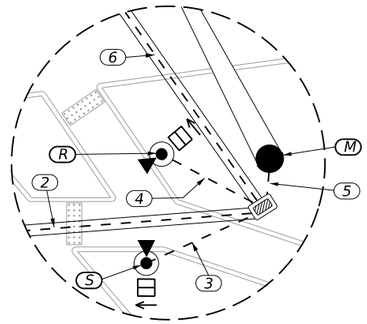
Q



DETAIL 4
(N.T.S.)



DETAIL 5
(N.T.S.)



DETAIL 6
(N.T.S.)

CALLOUTS:

- (A) PROP. 50 FT. MAST ARM POLE WITH STREET NAME SIGN AND PRESENCE RADAR DETECTOR.
- (B) PROP. 65 FT. MAST ARM POLE WITH LUMINAIRE, STREET NAME SIGN, PRESENCE RADAR DETECTOR AND ADVANCE RADAR DETECTOR.
- (C) PROP. 55 FT. MAST ARM POLE WITH LUMINAIRE, STREET NAME SIGN AND PRESENCE RADAR DETECTOR.
- (D, E, F, H, I, J) PROP. PEDESTAL POLE WITH PEDESTRIAN SIGNAL HEAD, SIGN AND PUSH BUTTON (APS UNIT).
- (G) PROP. PEDESTAL POLE WITH PEDESTRIAN SIGNAL HEADS (2 EA), SIGN AND PUSH BUTTON (APS UNIT).
- (K) PROP. 55 FT. MAST ARM POLE WITH LUMINAIRE, STREET NAME SIGN AND PRESENCE RADAR DETECTOR,
- (L) PROP. 55 FT. MAST ARM POLE WITH LUMINAIRE, STREET NAME SIGN, PRESENCE RADAR DETECTOR AND ADVANCE RADAR DETECTOR.
- (M) PROP. 55 FT. MAST ARM POLE WITH LUMINAIRE, STREET NAME SIGN AND PRESENCE RADAR DETECTOR.
- (N, O, P, R, S, T) PROP. PEDESTAL POLE WITH PEDESTRIAN SIGNAL HEAD, SIGN AND PUSH BUTTON (APS UNIT).
- (Q) PROP. PEDESTAL POLE WITH PEDESTRIAN SIGNAL HEADS (2 EA), SIGN AND PUSH BUTTON (APS UNIT).
- (U) PROP. FULL-ACTUATED CONTROLLER W/CABINET, GPS MODULE AND BATTERY BACK-UP (BBU)
- (V) PROP. ELECTRICAL SERVICE POLE TYPE D WITH METER, SERVICE (120/240 VOLT) AND SERVICE DISCONNECT
- (W) PROP. LUMINAIRE POLE

PROPOSED RADAR DETECTIONS SCHEDULE:

ADV1	DESIGNATED FOR SOUTHBOUND APPROACHING VEHICLES (SH 249)(FEEDER RD)
ADV2	DESIGNATED FOR NORTHBOUND APPROACHING VEHICLES (SH 249)(FEEDER RD)
PR1	DESIGNATED FOR SOUTHBOUND VEHICLES (SH 249)(FEEDER RD)
PR2	DESIGNATED FOR WESTBOUND VEHICLES (LOUETTA RD)
PR3	DESIGNATED FOR EASTBOUND VEHICLES (LOUETTA RD)
PR4	DESIGNATED FOR EASTBOUND VEHICLES (LOUETTA RD)
PR5	DESIGNATED FOR WESTBOUND VEHICLES (LOUETTA RD)
PR6	DESIGNATED FOR NORTHBOUND VEHICLES (SH 249)(FEEDER RD)

STATE OF TEXAS
ANUSHA POKHREL
151136
REGISTERED
PROFESSIONAL ENGINEER

4/29/2026

Texas Department of Transportation

SH 249
AT LOUETTA RD

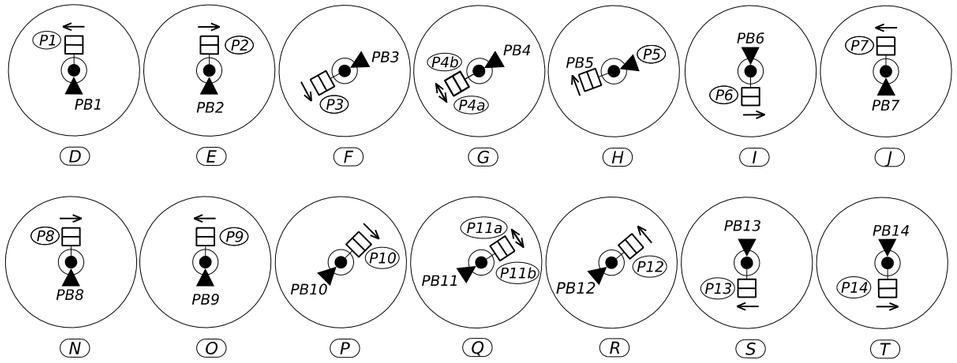
TRAFFIC SIGNAL
PROPOSED PLAN

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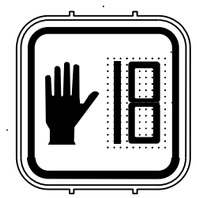
CONT	SECT	JOB	HIGHWAY
0720	03	153,ETC.	SH 249
DIST	COUNTY		SHEET NO.
HOU	HARRIS		34

DATE: 4/29/2026 9:27:51 AM
FILE: c:\txdot\pw_online\txdot03\hoang.tran1\d0985869\0720-03-154 SH 249 AT LOUETTA RD_OR.dgn

PROPOSED PEDESTRIAN SIGNAL HEAD
AND PUSH BUTTON ORIENTATION:



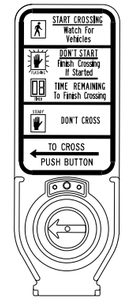
PROPOSED PEDESTRIAN
SIGNAL HEAD
(COUNTDOWN TYPE):



P1, P2, P3, P4a, P4b, P5, P6, P7
P8, P9, P10, P11a, P11b, P12, P13, P14

PROPOSED PEDESTRIAN SIGNS
AND PUSH BUTTONS:

R10-3eL



PB1, PB3, PB6, PB7,
PB9, PB12, PB14

R10-3eD



PB4, PB11

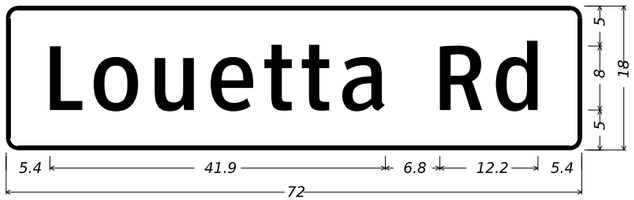
R10-3eR



PB2, PB5, PB8,
PB10, PB13

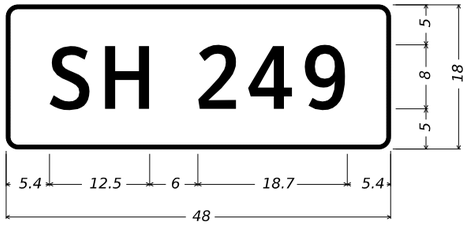
PROPOSED STREET SIGN:

S1



1.500" Radius, 0.500" Border, White on Green;
"Louetta Rd", ClearviewHwy-3-W;

S2



1.5" Radius, 0.5" Border, White on Green;
"SH", ClearviewHwy-3-W; "249", ClearviewHwy-3-W;

S3



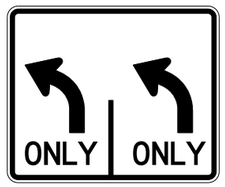
R6-1L 36x12;

S4



R6-1R 36x12;

S5



R3-8 MOD_36x30;

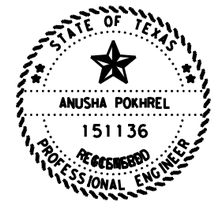
S6



R3-8R_30x30;

ELECTRICAL SERVICE DATA:

ELECTRICAL SERVICE NAME	CALLOUT	ELECTRICAL SERVICE DESCRIPTION SEE ED (5, 6, 7 & 8)-25	SERVICE CONDUIT SIZE (RMC)	SERVICE CONDUCTORS NO./SIZE	SAFETY SWITCH AMPS	MAIN CKT. BRK. POLE/AMP	TWO-POLE CONTACTOR AMPS	PANELBD./ LOAD CENTER AMP RATING (MIN)	CIRCUIT NO.	BRANCH CKT. BRK. POLE/AMPS	BRANCH CIRCUIT AMPS	kVA LOAD
SH 249 AT LOUETTA RD	V	ELEC SERV TY D (120/240)060(NS)SS(E)SP(O)	1-1/4"	3/#6	N/A	2P/60		100	TRF. SIG	1P/50	40	6.7
							30		LIGHTING	2P/20	4	



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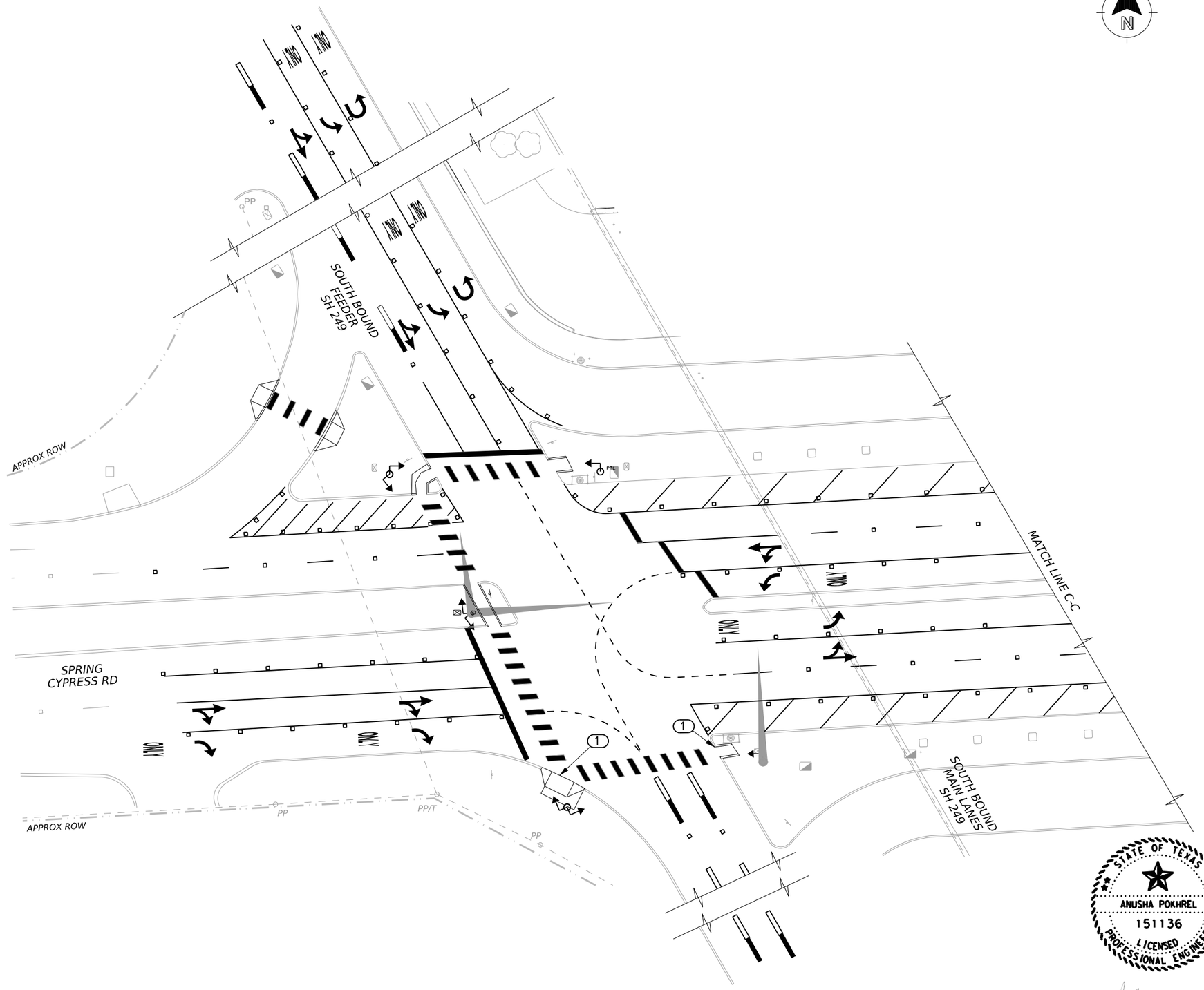
SH 249
AT LOUETTA RD

TRAFFIC SIGNAL
PROPOSED PLAN

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CONT	SECT	JOB	HIGHWAY
0720	03	153,ETC.	SH 249
DIST	COUNTY		SHEET NO.
HOU	HARRIS		35

DATE: 4/29/2026 9:27:51 AM
FILE: c:\txdot\pw_online\txdot51\hoang.tran1\d0985869\0720-03-154 SH 249 AT LOUETTA RD OR.dgn



0 20 40 FT

SCALE: 1" = 40'

STATE OF TEXAS
ANUSHA POKHREL
151136
LICENSED
PROFESSIONAL ENGINEER

4/29/2026

Texas Department of Transportation

SH 249
AT SPRING CYPRESS RD

PAVEMENT MARKINGS AND RAMPS EXISTING PLAN

©TxDOT 2026

SHEET 1 OF 2

CONT	SECT	JOB	HIGHWAY
0720	03	153,ETC.	SH 249
DIST	COUNTY		SHEET NO.
HOU	HARRIS		38

DATE: 4/29/2026 9:28:02 AM
FILE: c:\txdot\pw_online\txdot3\hoang.tran1\d0985869\0720-03-153 SH 249 at Spring Cypress Rd Pavement marking OR.dgn

CK
DW
CK
DW

APPROX ROW

NORTH BOUND
FEEDER
SH 249

PP/T

PP

PP

PP

PP

APPROX ROW

ONLY

SPRING
CYPRESS RD

AHEAD
LANE
TURN
LEFT

FOE

ONLY

ONLY

MATCH LINE C-C

NORTH BOUND
MAIN LANES
SH 249

ONLY

ONLY

ONLY

ONLY

APPROX ROW

N

0 20 40 FT

SCALE: 1" = 40'

STATE OF TEXAS
ANUSHA POKHREL
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PROFESSIONAL ENGINEER

4/29/2026

Texas Department of Transportation

SH 249
AT SPRING CYPRESS RD

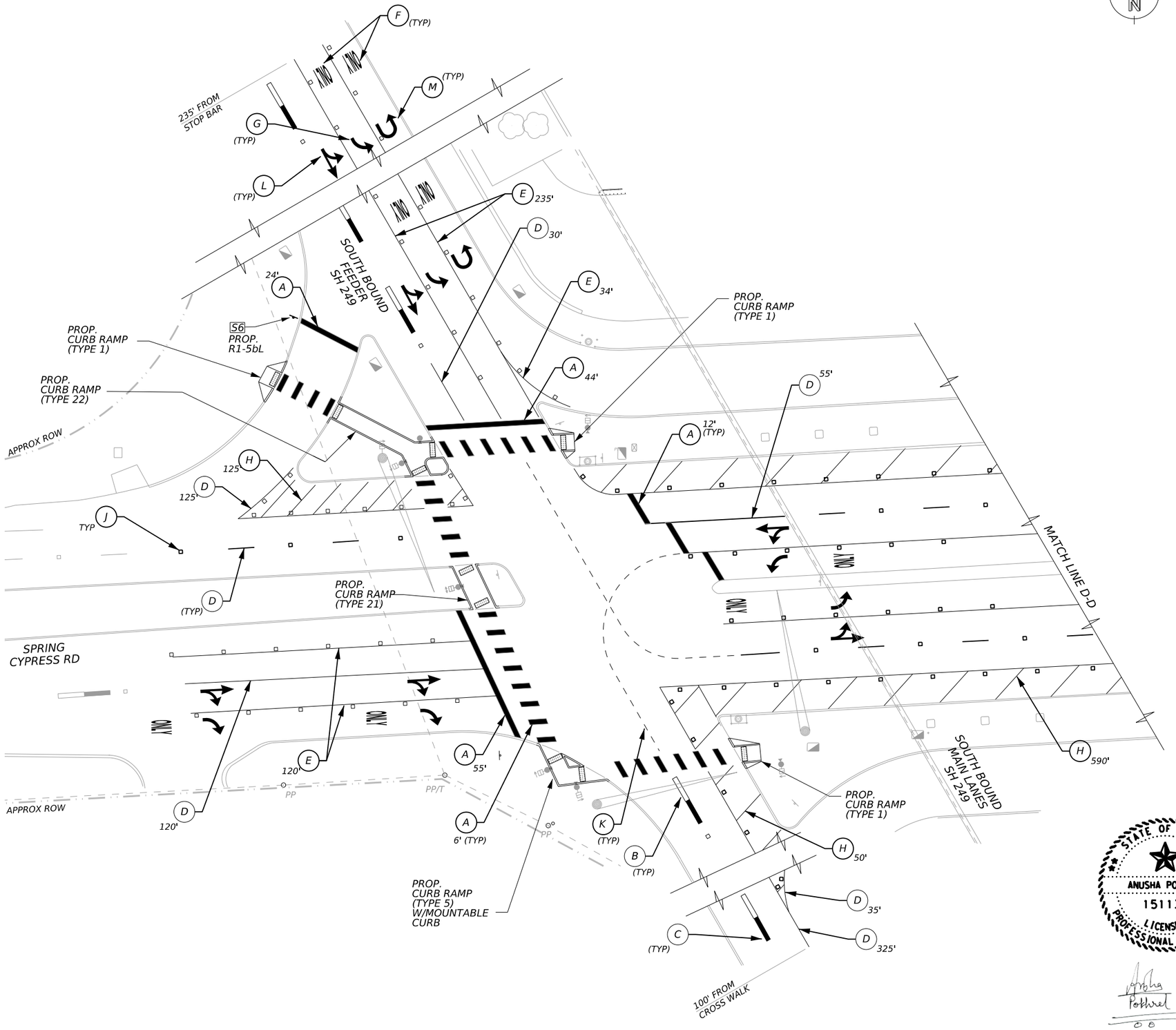
PAVEMENT MARKINGS AND RAMPS EXISTING PLAN

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SHEET 2 OF 2

CONT	SECT	JOB	HIGHWAY
0720	03	153,ETC.	SH 249
DIST	COUNTY		SHEET NO.
HOU	HARRIS		39

DATE: 4/29/2026 9:28:02 AM
FILE: c:\txdot\pw_online\txdot3\hoang.tran1\d0985869\0720-03-153 SH 249 at Spring Cypress Rd Pavement marking_OR.dgn



PROPOSED LEGEND

- (A) REFL PAV MRK TY I (W) 24" (SLD) (100MIL)
- (B) REFL PAV MRK TY I (W) 6" (BRK) (100MIL)
- (C) RE PV MRK TY I (BLACK) 6" (SHADOW) (100MIL)
- (D) REFL PAV MRK TY I (W) 6" (SLD) (100MIL)
- (E) REFL PAV MRK TY I (W) 8" (SLD) (100MIL)
- (F) PREFAB PM TY C (W)(WORD)
- (G) PREFAB PM TY C (W)(ARROW)
- (H) REFL PAV MRK TY I (W) 12" (SLD) (100MIL)
- (J) REFL PAV MRKR TY I-C
- (K) REFL PAV MRK TY I (W)6" (DOT)(100MIL)
- (L) PREFAB PM TY C (W)(DBL ARROW)
- (M) PREFAB PM TY C (W)(UTURN ARROW)

PROPOSED SMALL SIGN:

S6



R1-5BL_36X36;

0 20 40 FT

SCALE: 1" = 40'

STATE OF TEXAS
ANUSHA POKHREL
151136
LICENSED
PROFESSIONAL ENGINEER

4/29/2026

Texas Department of Transportation

SH 249
AT SPRING CYPRESS RD

PAVEMENT MARKINGS AND RAMPS PROPOSED PLAN

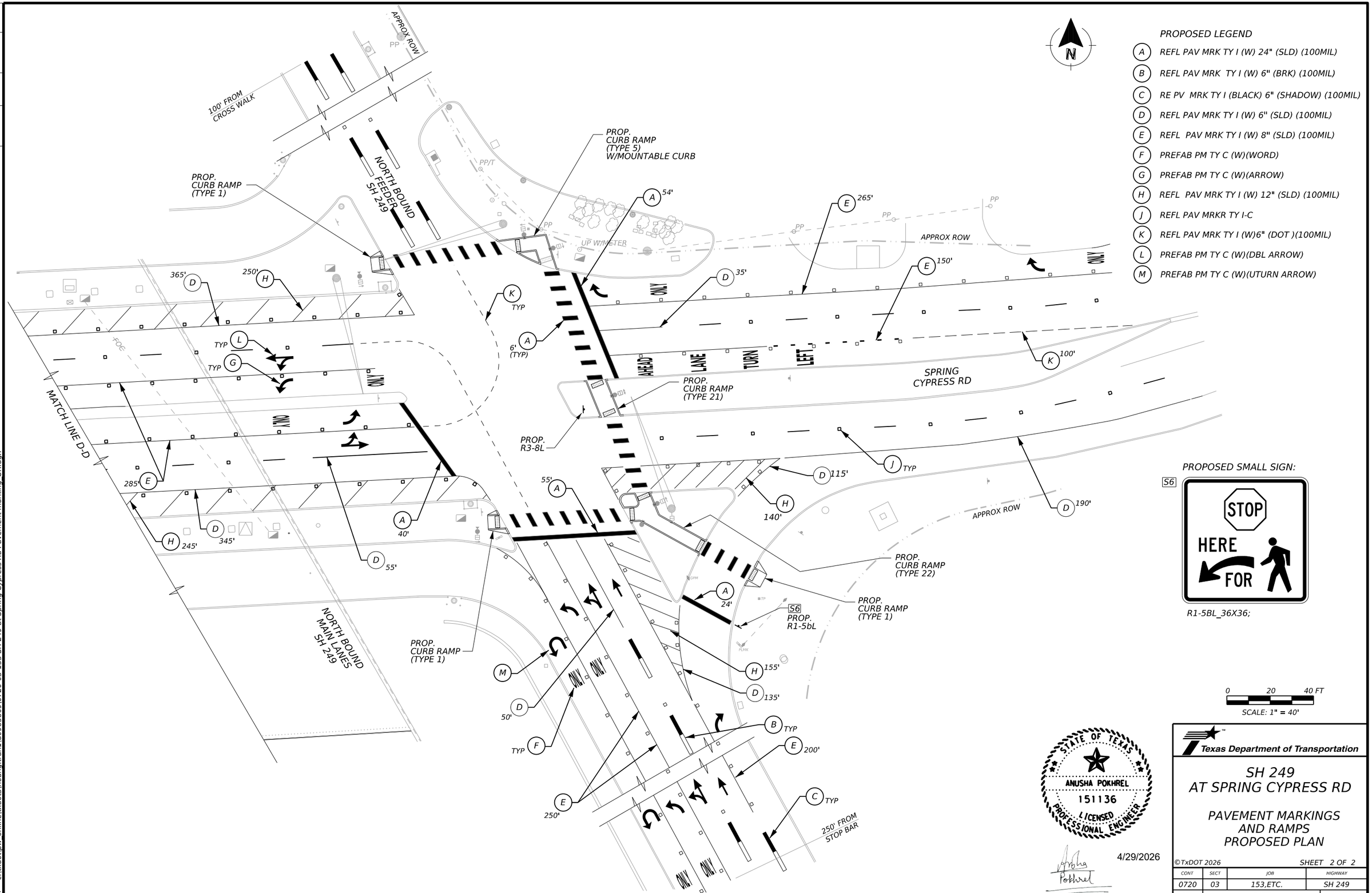
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SHEET 1 OF 2

CONT	SECT	JOB	HIGHWAY
0720	03	153,ETC.	SH 249
DIST	COUNTY		SHEET NO.
HOU	HARRIS		40

DATE: 4/29/2026 9:28:03 AM
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CK DW CK DW



DATE: 4/29/2026 9:28:03 AM
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4/29/2026

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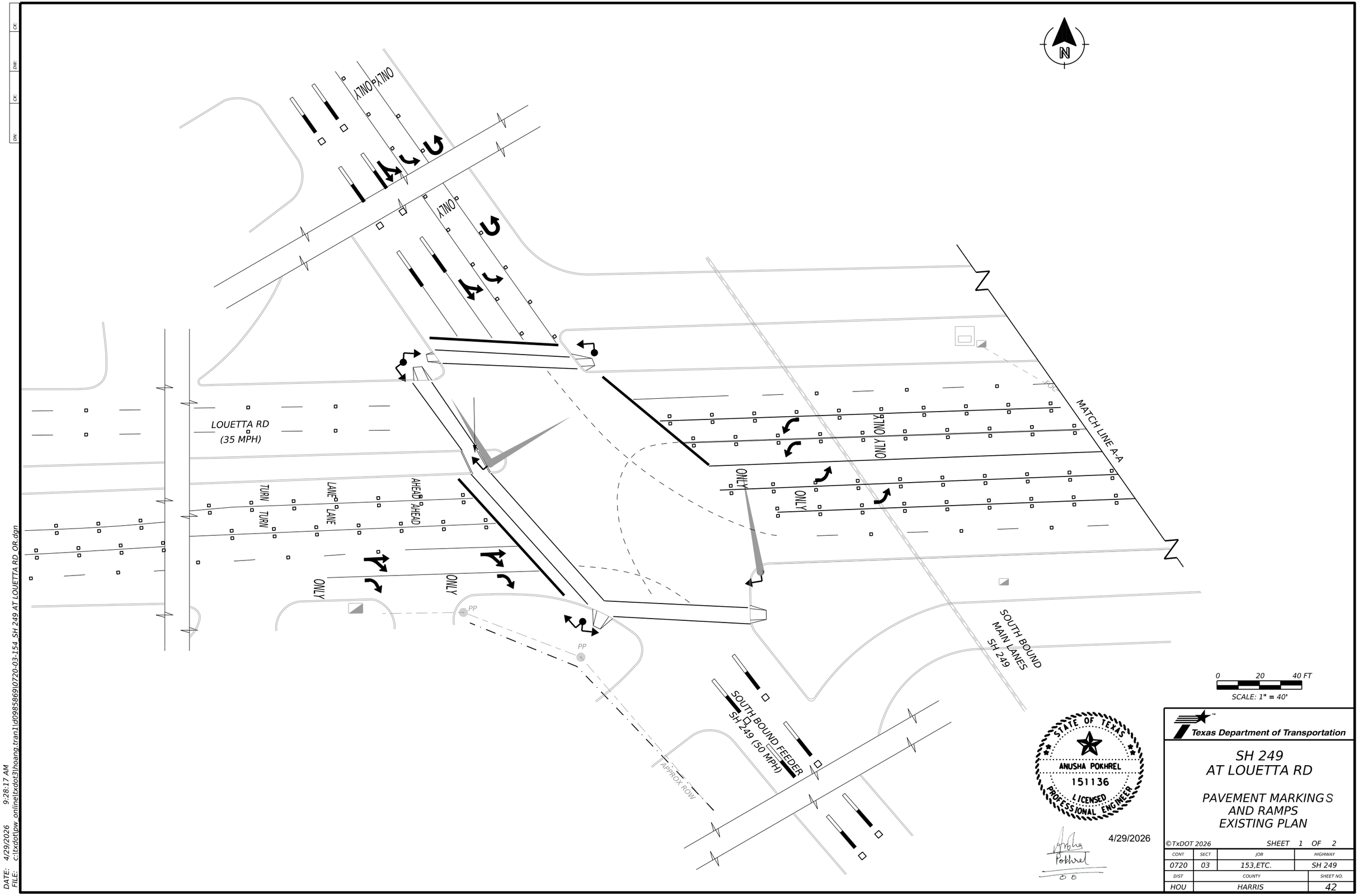
SH 249
AT SPRING CYPRESS RD

PAVEMENT MARKINGS AND RAMPS PROPOSED PLAN

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SHEET 2 OF 2

CONT	SECT	JOB	HIGHWAY
0720	03	153,ETC.	SH 249
DIST	COUNTY		SHEET NO.
HOU	HARRIS		41



DATE: 4/29/2026 9:28:17 AM
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4/29/2026

Texas Department of Transportation

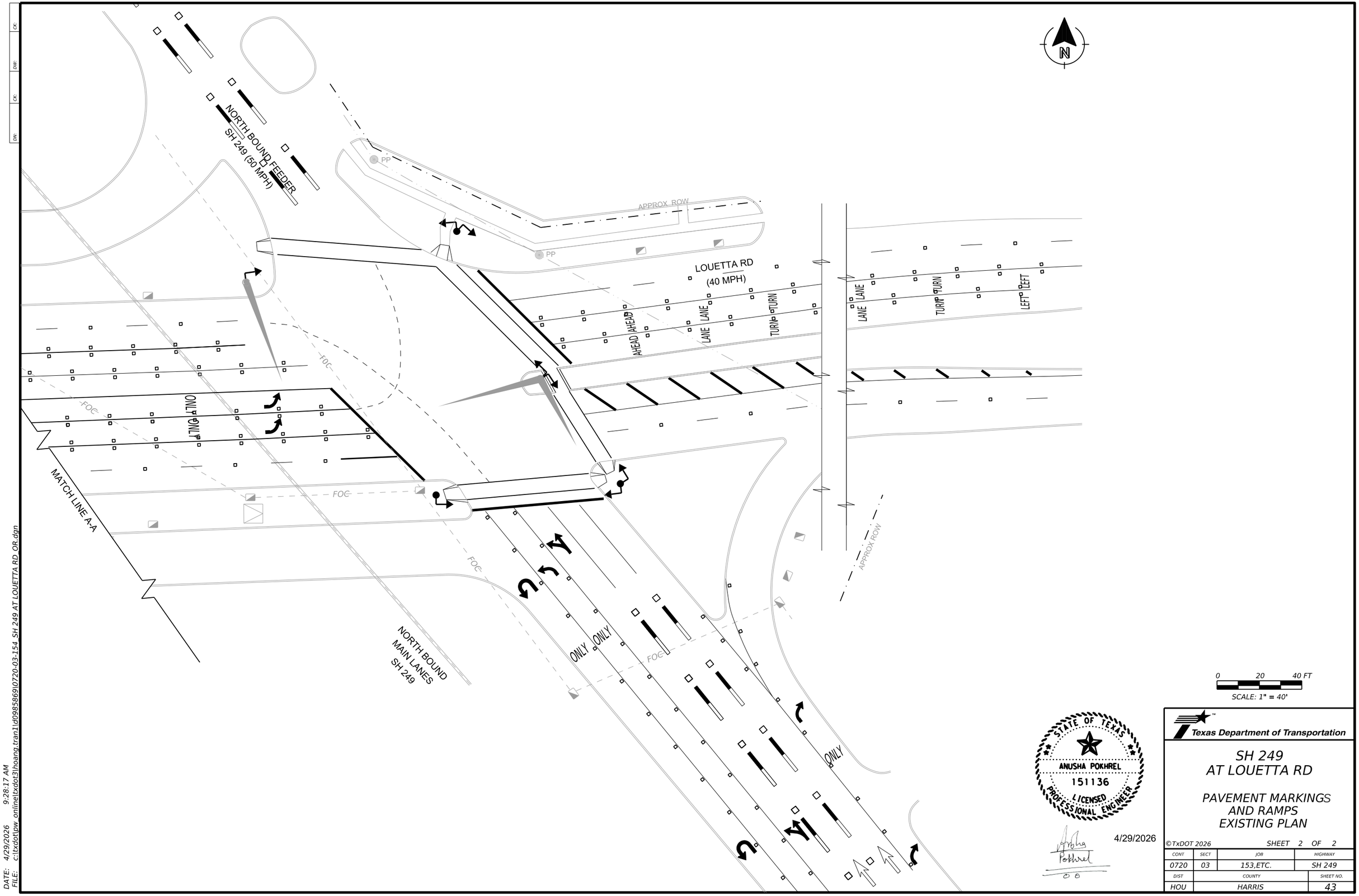
SH 249
AT LOUETTA RD

PAVEMENT MARKINGS AND RAMPS EXISTING PLAN

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SHEET 1 OF 2

CONT	SECT	JOB	HIGHWAY
0720	03	153,ETC.	SH 249
DIST	COUNTY		SHEET NO.
HOU	HARRIS		42



0 20 40 FT

SCALE: 1" = 40'

STATE OF TEXAS
ANUSHA POKHREL
151136
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PROFESSIONAL ENGINEER

4/29/2026

Texas Department of Transportation

SH 249
AT LOUETTA RD

PAVEMENT MARKINGS AND RAMPS EXISTING PLAN

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SHEET 2 OF 2

CONT	SECT	JOB	HIGHWAY
0720	03	153,ETC.	SH 249
DIST	COUNTY		SHEET NO.
HOU	HARRIS		43

DATE: 4/29/2026 9:28:17 AM
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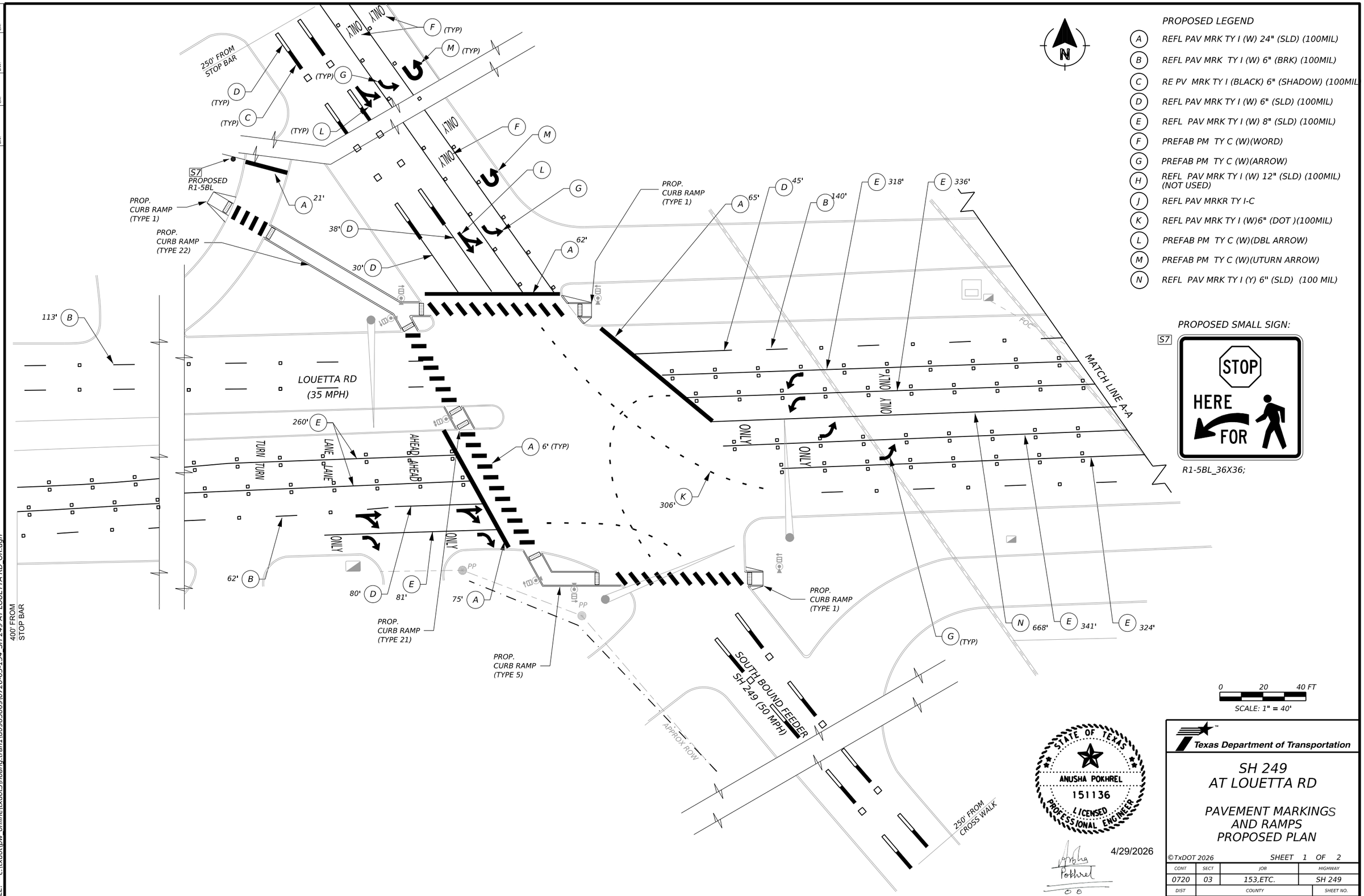
PROPOSED LEGEND

- (A) REFL PAV MRK TY I (W) 24" (SLD) (100MIL)
- (B) REFL PAV MRK TY I (W) 6" (BRK) (100MIL)
- (C) RE PV MRK TY I (BLACK) 6" (SHADOW) (100MIL)
- (D) REFL PAV MRK TY I (W) 6" (SLD) (100MIL)
- (E) REFL PAV MRK TY I (W) 8" (SLD) (100MIL)
- (F) PREFAB PM TY C (W)(WORD)
- (G) PREFAB PM TY C (W)(ARROW)
- (H) REFL PAV MRK TY I (W) 12" (SLD) (100MIL) (NOT USED)
- (J) REFL PAV MRKR TY I-C
- (K) REFL PAV MRK TY I (W)6" (DOT)(100MIL)
- (L) PREFAB PM TY C (W)(DBL ARROW)
- (M) PREFAB PM TY C (W)(UTURN ARROW)
- (N) REFL PAV MRK TY I (Y) 6" (SLD) (100 MIL)

PROPOSED SMALL SIGN:

S7

R1-5BL_36X36;



0 20 40 FT

SCALE: 1" = 40'

STATE OF TEXAS
ANUSHA POKHREL
151136
LICENSED
PROFESSIONAL ENGINEER

4/29/2026

Texas Department of Transportation

SH 249
AT LOUETTA RD

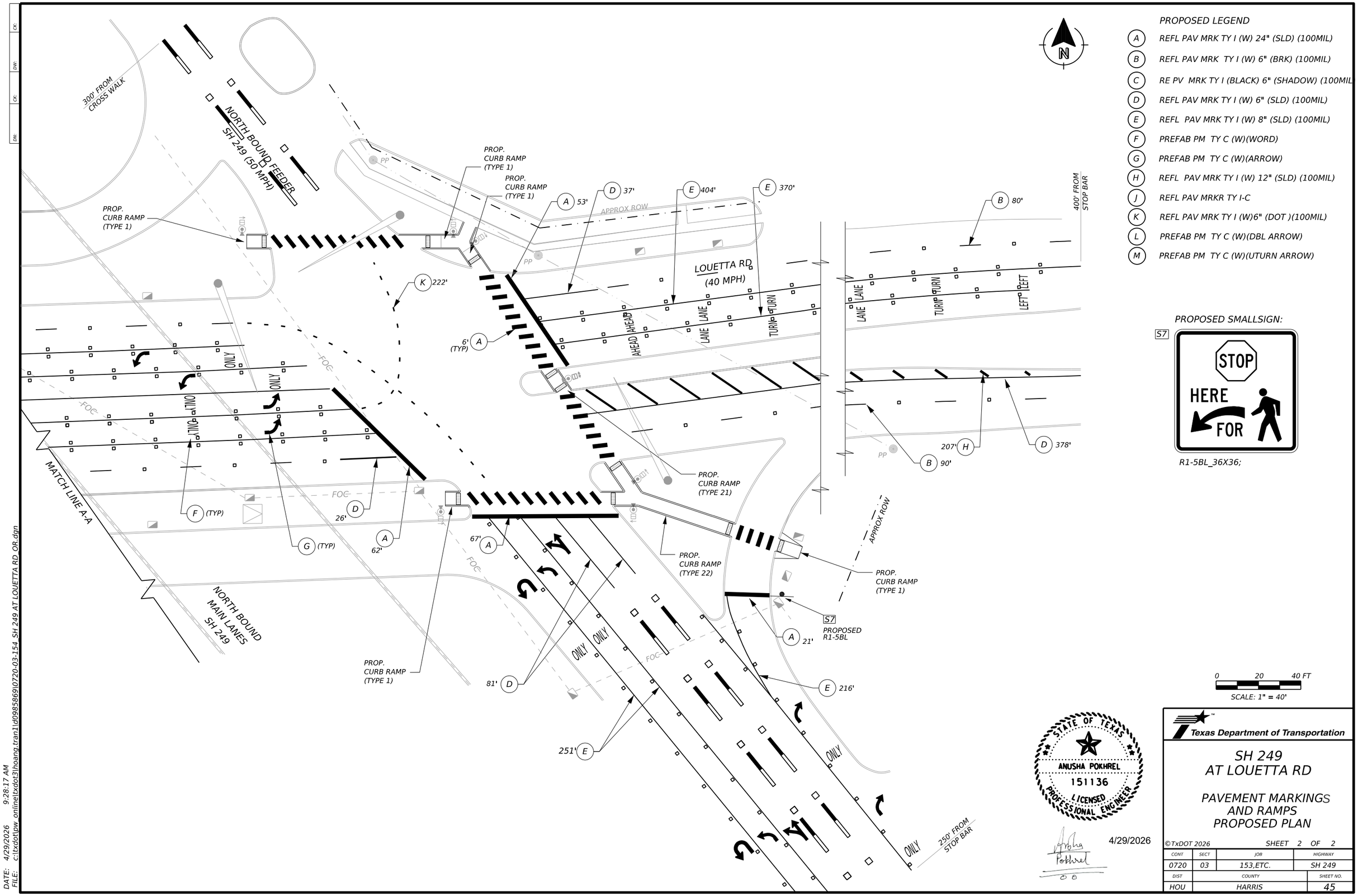
PAVEMENT MARKINGS AND RAMPS PROPOSED PLAN

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SHEET 1 OF 2

CONT	SECT	JOB	HIGHWAY
0720	03	153,ETC.	SH 249
DIST	COUNTY		SHEET NO.
HOU	HARRIS		44

DATE: 4/29/2026 9:28:17 AM
FILE: c:\txdot\pw_online\txdot03\hoang.tran1\d0985869\0720-03-154 SH 249 AT LOUETTA RD_OR.dgn



PROPOSED LEGEND

- (A) REFL PAV MRK TY I (W) 24" (SLD) (100MIL)
- (B) REFL PAV MRK TY I (W) 6" (BRK) (100MIL)
- (C) RE PV MRK TY I (BLACK) 6" (SHADOW) (100MIL)
- (D) REFL PAV MRK TY I (W) 6" (SLD) (100MIL)
- (E) REFL PAV MRK TY I (W) 8" (SLD) (100MIL)
- (F) PREFAB PM TY C (W)(WORD)
- (G) PREFAB PM TY C (W)(ARROW)
- (H) REFL PAV MRK TY I (W) 12" (SLD) (100MIL)
- (J) REFL PAV MRKR TY I-C
- (K) REFL PAV MRK TY I (W)6" (DOT)(100MIL)
- (L) PREFAB PM TY C (W)(DBL ARROW)
- (M) PREFAB PM TY C (W)(UTURN ARROW)

PROPOSED SMALLSIGN:

S7

R1-5BL_36X36;

0 20 40 FT

SCALE: 1" = 40'

STATE OF TEXAS
ANUSHA POKHREL
151136
LICENSED
PROFESSIONAL ENGINEER

4/29/2026

Texas Department of Transportation

SH 249 AT LOUETTA RD

PAVEMENT MARKINGS AND RAMPS PROPOSED PLAN

©TxDOT 2026

SHEET 2 OF 2

CONT	SECT	JOB	HIGHWAY
0720	03	153,ETC.	SH 249
DIST	COUNTY		SHEET NO.
HOU	HARRIS		45

DATE: 4/29/2026 9:28:17 AM
FILE: c:\txdot\pw_online\txdot3\hoang.tran1\d0985869\0720-03-154_SH 249 AT LOUETTA RD_OR.dgn

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4/23/2026 11:08:18 AM
c:\txdot\pw\online\txdot3\hoang.tran\d0986559\d067 lma-SpringCypress.dgn

Shipping Parts List

Ship each pole with the following attached: enlarged hand hole, pole cap, fixed arm connection bolts and washers, and any additional hardware listed in the table.

Nominal Arm Length	30' Poles with Luminaire: See note above plus: one (or two if ILSN attached) small hand hole, clamp-on simplex	24' Poles with ILSN: See note above plus one small hand hole	19.50' (Single Mast Arm) 20.25' (Dual Mast Arm) Poles with no Luminaire and no ILSN: See note above

Single Mast Arm

Lf ft.	Designation	Quantity	Designation	Quantity	Designation	Quantity
50	50L		50S		50	
55	55L	5	55S		55	
60	60L	1	60S		60	
65	65L		65S		65	

Dual Mast Arm

Lf ft.	Lc ft.	Designation	Quantity	Designation	Quantity	Designation	Quantity
50	20	5020L		5020S		5020	
	24	5024L		5024S		5024	
	28	5028L		5028S		5028	
	32	5032L		5032S		5032	
	36	5036L		5036S		5036	
	40	5040L		5040S		5040	
	44	5044L		5044S		5044	
55	20	5520L		5520S		5520	
	24	5524L		5524S		5524	
	28	5528L		5528S		5528	
	32	5532L		5532S		5532	
	36	5536L		5536S		5536	
	40	5540L		5540S		5540	
	44	5544L		5544S		5544	
60	20	6020L		6020S		6020	
	24	6024L		6024S		6024	
	28	6028L		6028S		6028	
	32	6032L		6032S		6032	
	36	6036L		6036S		6036	
	40	6040L		6040S		6040	
	44	6044L		6044S		6044	
65	20	6520L		6520S		6520	
	24	6524L		6524S		6524	
	28	6528L		6528S		6528	
	32	6532L		6532S		6532	
	36	6536L		6536S		6536	
	40	6540L		6540S		6540	
	44	6544L		6544S		6544	

Foundation Summary Table **

Location Ident.	Avg. N Blow/ft.	No. Each	Drill Shaft *** Length (feet) 48-A
SH 249 AT			
SPRING CYPRESS RD			
POLE A	10	1	22
POLE B	10	1	22
POLE C	10	1	22
POLE K	10	1	22
POLE L	10	1	22
POLE M	10	1	22
Total Drill Shaft Length			132

Notes
** Foundations may be listed separately or grouped according to similarity of location and type. Quantities are for the Contractor's information only.
*** Decimal lengths in Design Table are to allow interpolation for other penetrometer values. Round to nearest foot for entry into Summary Table.

Abbreviations
Lf= Fixed Arm Length
Lc= Clamp-on Arm Length (44' Max.)

Shipping Parts List

Traffic Signal Arms (Fixed Mount) (1 per pole)
Ship each arm with listed equipment attached

Nominal Arm Length	Type IV Arm (4 Signals) 3 Bracket Assembly and 4 CGB Connectors	
ft.	Designation	Quantity
50	50IV	
55	55IV	5
60	60IV	1
65	65IV	

Luminaire Arms (1 per 30' pole)

Nominal Arm Length	Quantity
8' Arm	6

ILSN Arm (Max. 2 per pole) Ship with clamps, bolts and washers

Nominal Arm Length	Quantity
7' Arm	
9' Arm	

Traffic Signal Arms (80 MPH Clamp-On Mount) (1 per pole)
Ship each arm with listed equipment attached

Nominal Arm Length	Type I Arm (1 Signal) 2 CGB connector and 1 clamp w/bolts and washers		Type II Arm (2 Signals) 1 Bracket Assembly and 3 CGB connectors, 1 clamp w/bolts and washers		Type III Arm (3 Signals) 2 Bracket Assembly and 4 CGB connectors, 1 clamp w/bolts and washers	
ft.	Designation	Quantity	Designation	Quantity	Designation	Quantity
20	20I-80					
24	24I-80		24II-80			
28	28I-80		28II-80			
32			32II-80		32III-80	
36			36II-80		36III-80	
40					40III-80	
44					44III-80	

Traffic Signal Arms (100 MPH Clamp-On Mount) (1 per pole)
Ship each arm with listed equipment attached

Nominal Arm	Type I Arm (1 Signal) 2 CGB connector and 1 clamp w/bolts and washers		Type II Arm (2 Signals) 1 Bracket Assembly and 3 CGB connectors, 1 clamp		Type III Arm (3 Signals) 2 Bracket Assembly and 4 CGB connectors, 1 clamp	
ft.	Designation	Quantity	Designation	Quantity	Designation	Quantity
20	20I-100					
24	24I-100		24II-100			
28	28I-100		28II-100			
32			32II-100		32III-100	
36			36II-100		36III-100	
40					40III-100	
44					44III-100	

Anchor Bolt Assemblies (1 per pole)

Anchor Bolt Diameter	Anchor Bolt Length	Quantity
2 1/2 "	5' - 3"	6

Each anchor bolt assembly consists of the following:
Top and bottom templates, 4 anchor bolts, 8 nuts, 8 flat washers and 4 nut anchor devices (type 2) per Standard Drawing "TS-FD"
Templates may be removed for shipment.

SH 249 AT
SPRING CYPRESS RD

STATE OF TEXAS
ANUSHA POKHREL
151136
LICENSED PROFESSIONAL ENGINEER

4/29/2026

Texas Department of Transportation
Traffic Operations Division

LONG MAST ARM ASSEMBLY PARTS LIST

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CONT	SECT	JOB	HIGHWAY
0720	03	153,ETC.	SH 249
DIST	COUNTY		SHEET NO.
HOU	HARRIS		67

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Shipping Parts List

Ship each pole with the following attached: enlarged hand hole, pole cap, fixed arm connection bolts and washers, and any additional hardware listed in the table.

Nominal Arm Length	30' Poles with Luminaire: See note above plus: one (or two if ILSN attached) small hand hole, clamp-on simplex		24' Poles with ILSN: See note above plus one small hand hole		19.50' (Single Mast Arm) 20.25' (Dual Mast Arm) Poles with no Luminaire and no ILSN: See note above	
Single Mast Arm						
Lf ft.	Designation	Quantity	Designation	Quantity	Designation	Quantity
50	50L		50S		50	1
55	55L	4	55S		55	
60	60L		60S		60	
65	65L		65S		65	1
Dual Mast Arm						
Lf ft. / Lc ft.	Designation	Quantity	Designation	Quantity	Designation	Quantity
50 / 20	5020L		5020S		5020	
50 / 24	5024L		5024S		5024	
50 / 28	5028L		5028S		5028	
50 / 32	5032L		5032S		5032	
50 / 36	5036L		5036S		5036	
50 / 40	5040L		5040S		5040	
50 / 44	5044L		5044S		5044	
55 / 20	5520L		5520S		5520	
55 / 24	5524L		5524S		5524	
55 / 28	5528L		5528S		5528	
55 / 32	5532L		5532S		5532	
55 / 36	5536L		5536S		5536	
55 / 40	5540L		5540S		5540	
55 / 44	5544L		5544S		5544	
60 / 20	6020L		6020S		6020	
60 / 24	6024L		6024S		6024	
60 / 28	6028L		6028S		6028	
60 / 32	6032L		6032S		6032	
60 / 36	6036L		6036S		6036	
60 / 40	6040L		6040S		6040	
60 / 44	6044L		6044S		6044	
65 / 20	6520L		6520S		6520	
65 / 24	6524L		6524S		6524	
65 / 28	6528L		6528S		6528	
65 / 32	6532L		6532S		6532	
65 / 36	6536L		6536S		6536	
65 / 40	6540L		6540S		6540	
65 / 44	6544L		6544S		6544	

Shipping Parts List

Traffic Signal Arms (Fixed Mount) (1 per pole)
Ship each arm with listed equipment attached

Nominal Arm Length	Type IV Arm (4 Signals) 3 Bracket Assembly and 4 CGB Connectors	
ft.	Designation	Quantity
50	50IV	1
55	55IV	4
60	60IV	
65	65IV	1

Luminaire Arms (1 per 30' pole)

Nominal Arm Length	Quantity
8' Arm	4

ILSN Arm (Max. 2 per pole) Ship with clamps, bolts and washers

Nominal Arm Length	Quantity
7' Arm	
9' Arm	

Traffic Signal Arms (80 MPH Clamp-On Mount) (1 per pole) Ship each arm with listed equipment attached

Nominal Arm Length	Type I Arm (1 Signal) 2 CGB connector and 1 clamp w/bolts and washers		Type II Arm (2 Signals) 1 Bracket Assembly and 3 CGB connectors, and 1 clamp w/bolts and washers		Type III Arm (3 Signals) 2 Bracket Assembly and 4 CGB connectors, and 1 clamp w/bolts and washers	
ft.	Designation	Quantity	Designation	Quantity	Designation	Quantity
20	20I-80					
24	24I-80		24II-80			
28	28I-80		28II-80			
32			32II-80		32III-80	
36			36II-80		36III-80	
40					40III-80	
44					44III-80	

Traffic Signal Arms (100 MPH Clamp-On Mount) (1 per pole) Ship each arm with listed equipment attached

Nominal Arm	Type I Arm (1 Signal) 2 CGB connector and 1 clamp w/bolts and washers		Type II Arm (2 Signals) 1 Bracket Assembly and 3 CGB connectors, and 1 clamp		Type III Arm (3 Signals) 2 Bracket Assembly and 4 CGB connectors, and 1 clamp	
ft.	Designation	Quantity	Designation	Quantity	Designation	Quantity
20	20I-100					
24	24I-100		24II-100			
28	28I-100		28II-100			
32			32II-100		32III-100	
36			36II-100		36III-100	
40					40III-100	
44					44III-100	

Anchor Bolt Assemblies (1 per pole)

Anchor Bolt Diameter	Anchor Bolt Length	Quantity
2 1/2 "	5' - 3"	6

Each anchor bolt assembly consists of the following: Top and bottom templates, 4 anchor bolts, 8 nuts, 8 flat washers and 4 nut anchor devices (type 2) per Standard Drawing "TS-FD". Templates may be removed for shipment.

Foundation Summary Table **

Location Ident.	Avg. N Blow/ft.	No. Each	Drill Shaft *** Length (feet) 48-A
SH 249 AT LOUETTA RD			
POLE A	10	1	22
POLE B	10	1	22
POLE C	10	1	22
POLE K	10	1	22
POLE L	10	1	22
POLE M	10	1	22
Total Drill Shaft Length			132

Notes

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SH 249 AT LOUETTA RD

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LONG MAST ARM ASSEMBLY PARTS LIST

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